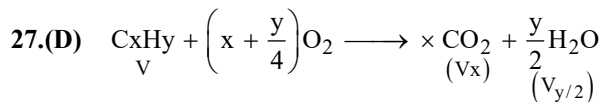


Solutions to Final Step-A | Chemistry

Stoichiometry - 1 & Redox Reactions

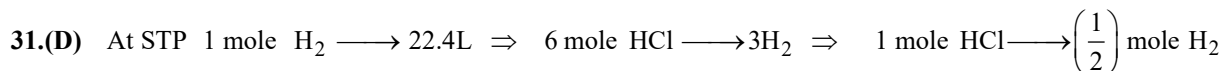
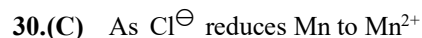
- 1.(D) As H_2O_2 is losing e^- s 2.(D) $\text{HClO} \xrightarrow{+1} \text{HCl} \xrightarrow{-1} + \text{HClO}_3 \xrightarrow{+5}$ 3.(A) $\text{H}_2\text{SO}_4 \xrightarrow{+6} \text{SO}_2 \xrightarrow{+4}$
 $\uparrow \quad \quad \quad \downarrow$
 $2e^- \text{ gain}$
- 4.(D) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+ : \text{Cr} = +3$
- 5.(C) $\text{MnO}_4^- \xrightarrow{+7} \text{Mn}^{2+} \xrightarrow{5e^-}$; $\text{MnO}_4^- \xrightarrow{+7} \text{MnO}_4^{2-} \xrightarrow{1e^-}$; $\text{MnO}_4^- \xrightarrow{+7} \text{MnO}_2 \xrightarrow{3e^-}$; $\text{MnO}_4^- \xrightarrow{+7} \text{Mn}_2\text{O}_3 \xrightarrow{4e^-}$
- 6.(C) Anode - where oxidation takes place 7.(C) $T(^{\circ}\text{F}) = 32 + \frac{9}{5}T(^{\circ}\text{C})$
- 8.(D) $\text{He} \longrightarrow 1 \text{ mole}$; $\text{Na} \longrightarrow 2 \text{ moles}$; $\text{Ca} \longrightarrow 10^{-2} \text{ moles}$; $\text{He} \longrightarrow 3 \text{ moles}$
- 9.(A) $(\text{H}_2\text{SO}_4) \text{ moles} = 0.1 \times 0.02 = 2 \times 10^{-3} \Rightarrow 2 \times 10^{-3} (\text{H}_2\text{SO}_4) \equiv 2 \times 10^{-3} \times N_A \text{ molecules}$
 Moles of = $12.044 \times 10^{20} \text{ molecules}$
- 10.(A) $(3.12 \times 1.5)_g$ to be one decimal as minimum significant figure has one digit after decimal.
- 11.(C) Example: $\text{H}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \longrightarrow \text{H}_2\text{O}(\ell)$ 12.(B) Typical case of Law of multiple proportions.
- 13.(AD) $\text{O}_2 \longrightarrow 32g$ For 1 mol there are N_A molecules
- 14.(AB) $[\text{NaOH}] \longrightarrow M = \frac{20}{40} \times \frac{1000}{200} = \frac{5}{2} M$; $[\text{KOH}] \longrightarrow M = 0.5 \times \frac{1000}{200} = \frac{5}{2} M$
- 15.(CD) Refer formula 16.(BD) Refer theory
- 17.(A) As bromine case more change in oxidation state
- 18.(A) $\text{NaH} \longrightarrow \text{Oxidation State} \rightarrow (-1)$ as compared to iodine
- 19.(B) $(\text{NH}_4^+) \quad (\text{NO}_3^-)$
 $\downarrow \quad \quad \downarrow$
 $-3 \quad \quad +5$
- 20.(D) $\text{Mn} \longrightarrow 3d^5 4s^2 (+7)$
- 21.(D) $\text{NO}_2 \xrightarrow{+4} \text{NO}_2^- \xrightarrow{+3}$ $\text{NO}_2 \xrightarrow{+4} \text{NO}_3^- \xrightarrow{+5}$
 $\uparrow \quad \quad \quad \uparrow$ $\uparrow \quad \quad \quad \uparrow$
 reduction oxidation
- 22.(ABC) $2\text{KClO}_3 \xrightarrow{+1 \ +5 \ -2} 2\text{KCl} \xrightarrow{+1 \ -1} + 3\text{O}_2 \xrightarrow{0}$
 $\uparrow \quad \quad \quad \uparrow$
 reduction oxidation
- 23.(CD) $\text{Zn} \xrightarrow{0} + 2\text{HCl} \xrightarrow{+1} \text{ZnCl}_2 \xrightarrow{+2} + \text{H}_2 \xrightarrow{0}$
 $\uparrow \quad \quad \quad \uparrow$
 Oxidant reductant
- 24.(CD) C : Vanadium, D : Phosphorous
- 25.(B) As oxidizing nature depends on electron gain enthalpy and hydration energy.
- 26.(C) Mn acts as oxidizing agent and oxidation state changes from +7 to +2.



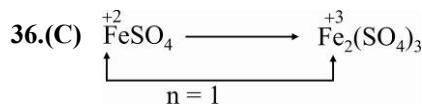
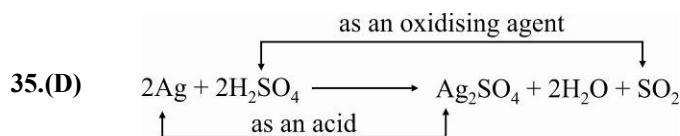
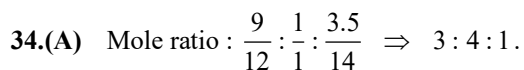
$$n_{H_2O} = \left(\frac{0.72}{18}\right)n_{CO_2} = \left(\frac{3.08}{44}\right) \Rightarrow \frac{n_{CO_2}}{n_{H_2O}} = \frac{3.08}{44} \times \frac{18}{0.72} \Rightarrow \frac{x}{y/2} = \frac{7}{4} \Rightarrow \frac{x}{y} = \frac{7}{8}$$

28.(D) $Volume = \left[\frac{\overset{\text{Total Mass}}{1120}}{1.15} \right] \longrightarrow \text{ml M} = \frac{120}{60} \times \left(\frac{1000 \times 1.15}{1120} \right) = 2.05M$

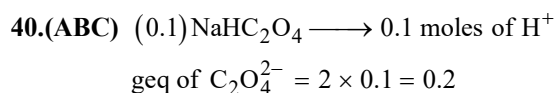
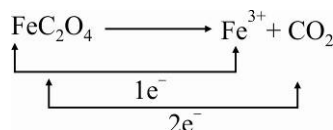
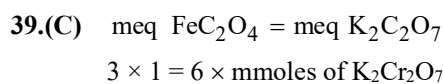
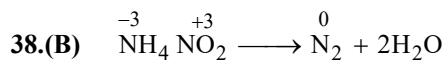
29.(C) $\frac{g}{(294/6)} = 0.75 \times 0.6 \times 1 \Rightarrow g = 22.05g$



32.(B) Mass will remain constant

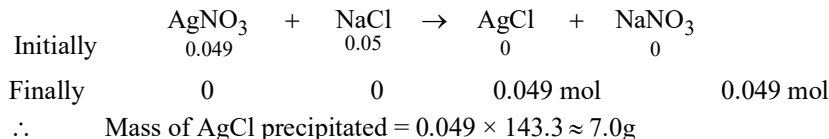


37.(C) Refer structure from module



41.(B) 16.9% solution of $AgNO_3$ means 16.9 g of $AgNO_3$ in 100 mL of solution.
 16.9% of $AgNO_3$ in 100 mL solution $\equiv$ 8.45 g of $AgNO_3$ in 50 mL $\equiv$ 0.049 mol
 Similarly, 5.8% of $NaCl \equiv$ 2.9 g of $NaCl$ in 50 mL solution $\equiv$ 0.05 mol.

The reaction can be represented as :



42.(A) Mass of 1 mol (6.022×10^{23} atoms) of carbon = 12 g.
 If Avogadro number is changed to 6.022×10^{20} atoms then mass of 1 mol of carbon

$$= \frac{12 \times 6.022 \times 10^{20}}{6.022 \times 10^{23}} = 12 \times 10^{-3} g$$

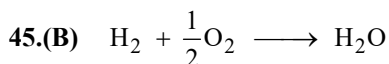
- 43.(C) According to Avogadro's hypothesis, ratio of the volumes of gases will be equal to the ratio of their moles.

$$n_{H_2} = \frac{g}{2} : n_{O_2} = \frac{g}{32} ; n_{CH_4} = \frac{g}{16}$$

So, the ratio is $\frac{g}{2} : \frac{g}{32} : \frac{g}{16}$ or 16 : 1 : 2 .

44.(D) Moles of urea = $\frac{6.02 \times 10^{20}}{6.02 \times 10^{23}} = 0.001$

$$\text{Concentration of solution} = \frac{0.001}{100} \times 1000 = 0.01 \text{ M}$$



1 mol 0.5 mol 1 mol

10 g of $H_2 = 5$ mol and 64 g of $O_2 = 2$ mol

$\therefore$ In this reaction, oxygen is the limiting reagent hence amount of H_2O produced depends on that of O_2 .

Since 0.5 mol of O_2 gives 1 mol H_2O

$\therefore$ 2 mol of O_2 will give 4 mol H_2O

46.(D) Average isotopic mass of X = $\frac{200 \times 90 + 199 \times 8 + 202 \times 2}{90 + 8 + 2} = \frac{18000 + 1592 + 404}{100} = 199.96 \text{ amu} \sim 200 \text{ amu}$

47.(A) At STP: 22.4 L $H_2 = 6.023 \times 10^{23}$ molecules

$$15 \text{ L } H_2 = \frac{6.023 \times 10^{23} \times 15}{22.4} = 4.033 \times 10^{23}$$

$$5 \text{ L } N_2 = \frac{6.023 \times 10^{23} \times 5}{22.4} = 1.344 \times 10^{23}$$

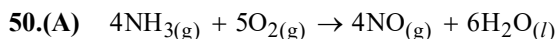
$$0.5 \text{ g } H_2 = \frac{6.023 \times 10^{23} \times 0.5}{2} = 1.505 \times 10^{23}$$

$$10 \text{ g of } O_2 = \frac{6.023 \times 10^{23} \times 10}{32} = 1.882 \times 10^{23}$$

- 48.(D) Zeros placed left to the number are never significant, therefore the number of significant figures for the numbers. 161 cm, 0.161 and 0.0161 are same, i.e. 3.

49.(A) Amount of iron in one molecule (in gms) = $\frac{67200}{100} \times 0.334 = 224.45 \text{ amu}$

$$\text{Number of iron atoms in one molecule of hemoglobin} = \frac{224.45}{56} = 4 .$$



4 mol $NH_{3(g)} \equiv 5$ moles $O_{2(g)} \equiv 4$ moles $NO_{(g)} \equiv 6$ moles $H_2O_{(l)}$

Observe that O_2 is limiting reagent, so it will be consumed totally leaving behind NH_3 .

51.(A) 1 mol CCl_4 vapour = $12 + 4 \times 35.5 = 154 \text{ gm}$

$$\text{At S.T.P.: Density of } CCl_4 \text{ vapour} = \frac{154}{22.4} \text{ g L}^{-1} = 6.875 \text{ g L}^{-1} .$$



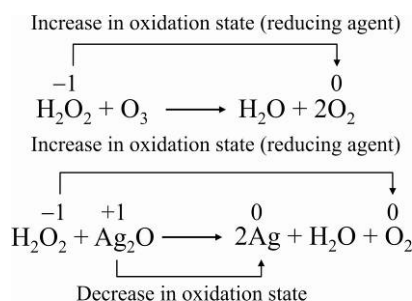
1 gmeq of Mg^{2+} requires 1 gmeq of soda.

$$12.00 \text{ mg} \equiv 12.00 \times 10^{-3} \text{ g} \equiv \frac{12.00 \times 10^{-3}}{24/2} \times 1000 = 1 \text{ meq of } \text{Mg}^{2+}$$

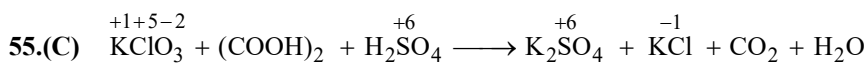
$\Rightarrow$ 1 meq of soda is required.

53.(C) Both FeCl_2 and SnCl_2 are reducing agents with lower oxidation numbers.

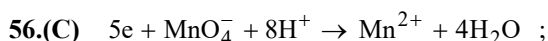
54.(C)



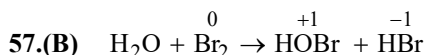
H_2O_2 acts as reducing agent in all those reactions in which O_2 is evolved..



Maximum change in oxidation number of chlorine, i.e., from +5 to -1.



$$\text{gmeq of } \text{MnO}_4^- = \text{gmeq of } \text{FeC}_2\text{O}_4 \Rightarrow 5n = 3 \times 1 \Rightarrow n = \frac{3}{5} = 0.6$$



In the above reaction the oxidation number of Br_2 increases from zero (in Br_2) to +1 (in HOBr) and decreases from zero (in Br_2) to -1 (in HBr). Thus Br_2 is oxidised as well as reduced and hence it is a disproportionation redox reaction.

58.(B) meq of $\text{NaOCl} = \text{meq of } \text{NaCrO}_2$.



n-factor of $\text{NaOCl} = 2$

n-factor of $\text{NaCrO}_2 = 3$

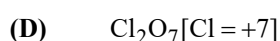
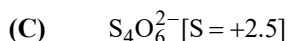
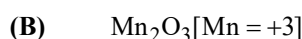
$$\Rightarrow 2 \times 0.15 \times V = 3 \times 0.2 \times 20 \Rightarrow V = 40 \text{ mL}$$

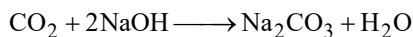
59.(D) meq of $\text{H}_2\text{O}_2 = \text{meq of } \text{MnO}_4^- / \text{H}^+$

$$\Rightarrow N \times 10 = \frac{1}{0.56} \times 10 \Rightarrow N(\text{H}_2\text{O}_2) = \frac{1}{0.56}$$

$$\text{Volume strength of } \text{H}_2\text{O}_2 ; 1 \text{ N } \text{H}_2\text{O}_2 \equiv 5.6 \times \frac{1}{0.56} = 10 \text{ vol}$$

60.(C) Find oxidation state of each element.





$$\begin{array}{ccc} 90 \text{ m mole} & 500 \text{ mmole} & 0 \\ 0 & 500-180 & 90 \\ & = 320 \text{ mmole} & \end{array}$$

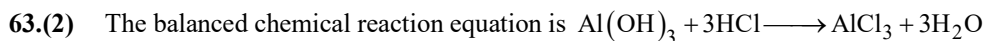
320 mmole of NaOH and 90 mmole of Na_2CO_3 will need 410 mmole of HCl. Since HCl is 0.5 M

$\Rightarrow$ Volume required is 820 ml

62. (4) $\text{Mol of O}_2 = \frac{44.8}{22.4} = 2.0$ $\text{Mol of CH}_3\text{OH} = \frac{64}{32} = 2.0$

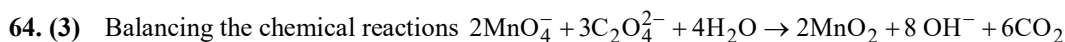
$\therefore$ 2 mol CH_3OH require 3 mol O_2 , hence O_2 is a limiting reagent.

Moles of CO_2 formed = $\frac{2}{3} \times 2 = 1.33$



Now, mol of HCl produced = 0.08×3

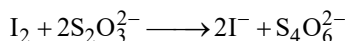
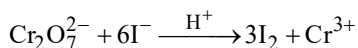
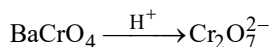
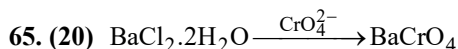
$\Rightarrow$ mol. of $\text{Al}(\text{OH})_3$ needed = $0.08 \Rightarrow$ mass of $\text{Al}(\text{OH})_3$ needed = $0.08 \times 78 = 6.24$



Since,

$\Rightarrow (2 + 6) - (2 + 3)$

$\Rightarrow 3$



$\therefore$ milli moles of $\text{I}_2 = \frac{90 \times 0.2}{2} = 9$

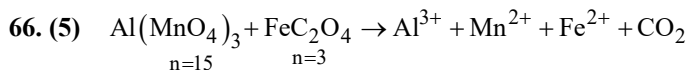
milli moles of $\text{Cr}_2\text{O}_7^{2-} = 3$

milli moles of $\text{BaCl}_2 \cdot 2\text{H}_2\text{O} = 6$

Amount of $\text{BaCl}_2 \cdot 2\text{H}_2\text{O} = 6 \times 10^{-3} \times 244 = 1.464$

% purity = $\frac{1.464}{7.32} \times 100$

= 20%

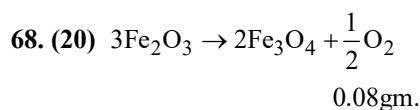


67. (2) Normality of acid = $\frac{39}{82/n \times 1}$

Normality of NaOH = $\frac{40}{40} \times \frac{1000}{1000} = 1$

$$\text{Now, meq. of acid} = \text{meq. of NaOH} \Rightarrow \frac{39n}{82} \times 100 = 1 \times 95$$

$$n = 2$$



$$w = \frac{0.08}{32} \times 2 \times 3 \times 160$$

$$= 2.4\text{gm}$$

$$\therefore \% \text{SiO}_2 = \frac{0.6}{3} \times 100$$

$$= 20\%$$

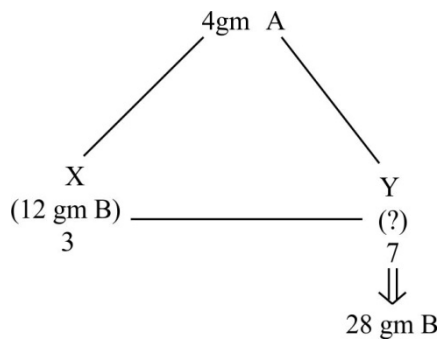
69. (24) $\frac{a \times 16 + b \times 28}{a + b} = 20$

$$\Rightarrow a = ab$$

$$\frac{b \times 16 + a \times 28}{a + b} = ?$$

$$\Rightarrow M = 24$$

70. (56) 56 gm



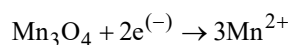
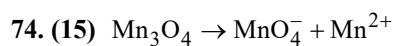
$$\therefore 8 \text{ gm A} \rightarrow 56 \text{ gm B}$$

71. (10) $\frac{a}{72} \times 1 + \frac{b}{160} \times 2 = \frac{7}{56} \quad \dots(1)$

$$\frac{a}{72} \times 1 + \frac{b}{160} \times 3 = \frac{2.5}{16} \quad \dots(2)$$

72. (1) $\frac{\frac{500 \times 0.2}{1000} + \frac{17.1 \times 3}{342}}{0.6 \times 5} = \frac{1}{12} \text{ M.}$

73. (1) $\frac{17.4}{87} = \frac{12.6}{110 + 16y} \times 2 \quad \Rightarrow \quad y = 1$



$$n_{\text{eff}} = \frac{2 \times 13}{2 + 13} = \frac{26}{15} \Rightarrow E = \frac{M}{26} \times 15$$

75. (3) 4m mol KI $\longrightarrow$ 2 m mole $e^{(-)}$

5m mol KI $\longrightarrow \frac{2}{4} \times 5$ m mole $e^{(-)}$

Atomic Structure

1.(ABD) Option (C) is postulate of Bohr's model

2.(B) Cu has $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ ($z = 29$)

3.(ABCD) Refer theory

4.(ACD) Refer theory

5.(A) Plum pudding model

6.(D) Same atomic mass but different atomic numbers

7.(D) $3p \rightarrow n - \ell - 1$ (radial nodes) = $3 - 1 - 1 = 1$

8.(C) $4d \rightarrow \ell$ (angular nodes)

9.(B) Refer theory

10.(B) Fe^{3+} , Mn^{2+} have same no. of electrons

11.(D) Pauli's Exclusion Principal

12.(CD) For an atom ${}_Z^AX$; A : Mass number, Z : Atomic Number ; For isotopes Z is same
 $\downarrow$
symbol

13.(AD) Energy is governed by $(n + \ell)$ rule

14.(BC) As $n = \ell$ (is not allowed) ; $n < \ell$ (is not allowed) 15.(AC) Both pairs have $10e^{-}$ s

16.(A) Isotopes have same number of electrons

17.(B) (R) is not correct explanation of (A)

18.(A) $\frac{hc}{\lambda} = 2.18 \times 10^{-18} \left(\frac{1}{1^2} - \frac{1}{2^2} \right) (1)^2$

$$\frac{6.625 \times 10^{-34} \times 3 \times 10^8}{\lambda} = 2.18 \times \frac{3}{4} \times 10^{-18} \Rightarrow \lambda = \left(\frac{6.626 \times 4}{2.18} \right) \times 10^{-8} = 1.214 \times 10^{-7} \text{ m}$$

19.(B) Apply $(n + \ell)$ rule : For $n_1 + \ell_1 = n_2 + \ell_2$
 $n_2 > n_1$

The electron with n_2 principal quainter number has more energy.

20.(C) Apply $\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \Rightarrow \frac{1}{355} = \frac{1}{680} + \frac{1}{\lambda_2} \Rightarrow \frac{680 - 355}{680 \times 355} = (\lambda_2)^{-1} \Rightarrow \lambda_2 = \frac{680 \times 355}{325} = 742.76 \text{ nm}$

21.(B) For Helium ion, $E \propto \frac{Z^2}{n^2} \propto (2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$ for hydrogen atom

22.(D) $\frac{hc}{\lambda} \times N_A = 242 \times 10^3 \Rightarrow \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{242 \times 10^3} = 494 \text{ nm}$

23.(B) $[He^+] = -19.6 \times 10^{-18} = E_n \left(\frac{Z^2}{1^2} \right) = E_n \left(\frac{2^2}{1^2} \right) \Rightarrow E_n = -4.9 \times 10^{-18}$

$$[Li^{2+}] = -4.9 \times 10^{-18} \frac{(3)^2}{1^2} = -4.41 \times 10^{-17} \text{ J atom}^{-1}$$

24.(C) $\Delta x \cdot \Delta p = \frac{h}{4\pi}$

$$\Delta x \cdot (9.1 \times 10^{-31} \times (0.005 \times 10^{-2}) 600) \geq \frac{6.626 \times 10^{-34}}{4\pi} \Delta x \cdot (9.1 \times 10^{-33} \times 3) \geq \frac{6.626 \times 10^{-34}}{4\pi}$$

$$\Delta x \geq \frac{6.626 \times 10^{-1}}{4\pi \times 3 \times 9.1} \Rightarrow \Delta x \geq 1.92 \times 10^{-3} \text{ m}$$

25.(B) $n + \ell$ is greatest

26.(A) As neutron emission only change A not Z.

27.(C) $\Delta v = (300 \times 0.001 \times 10^{-2} \text{ m/s})$

$$\Delta x \cdot (9.1 \times 10^{-31} \times 3 \times 10^{-3}) \geq \frac{6.63 \times 10^{-34}}{4\pi} \Rightarrow \Delta x \geq \left(\frac{6.63}{4\pi \times 9.1 \times 3} \right) \Rightarrow \Delta x \geq 1.92 \times 10^{-2} \text{ m}$$

28.(D) $mvr = \frac{nh}{2\pi} = \frac{5}{2} \frac{h}{\pi} = \frac{2.5h}{\pi}$

29.(A) In single electron system energy is governed by principal quantum numbers only

30.(BC) Refer theory

31.(C) For 4f orbital $n = 4, \ell = 3$

32.(B) $\text{Cr} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

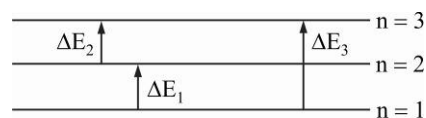
33.(B) Ist line $3 \rightarrow 2$, IInd line $4 \rightarrow 2$, IIIRD line $5 \rightarrow 2$

34.(D) Fe : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$; Fe^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$

35.(B) $\ell = 0$ (s) Angular momentum = 0

36.(C) Second excited state ($n = 3$) : $E(n = 3) = -\frac{13.6}{(3)^2} (1)^2 = -1.51 \text{ eV}$

37.(A) $\Delta E_1 + \Delta E_2 = \Delta E_3 \Rightarrow h\nu_1 + h\nu_2 = h\nu_3 \Rightarrow \nu_1 + \nu_2 = \nu_3$



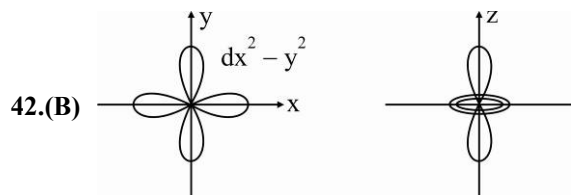
38.(C) Number of waves in nth orbit = n

39.(D) K.E. = $h\nu - h\nu_0$

$$h\nu = 1.5 h\nu_0 \Rightarrow \text{K.E.} = 0.5h\nu_0 \Rightarrow h\nu_0 = 13.6 \Rightarrow \text{K.E.} = 13.6 \times 0.5 = 6.8 \text{ eV}$$

40.(D) $E_{n_1 \rightarrow n_2} = 13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

41.(D) Total energy = $-13.6(2)^2$; P.E. = $2T.E. = -13.6 \times 2 \times (2)^2 = -108.8 \text{ eV}$



43.(B) $\text{Ti}(22) 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2 = \sqrt{n(n+2)} = \sqrt{2 \times 4} = \sqrt{8} \text{ B.M.}$

44.(B) Refer theory

45.(C) $a_0 \longrightarrow$ Bohr's radius $r = a_0 \left(\frac{n^2}{z} \right) \left\{ \begin{matrix} n=2 \\ z=1 \end{matrix} \right\}$

$$mvr = \frac{nh}{2\pi} \Rightarrow mva_0 \frac{n^2}{z} = \frac{nh}{2\pi} \Rightarrow mv = \left(\frac{hz}{2\pi a_0 n} \right) (z=1)$$

$$mv = \left(\frac{h}{2\pi a_0 n} \right) \Rightarrow mv = \left(\frac{h}{4\pi a_0} \right) (n=2) \Rightarrow \frac{(mv)^2}{2m} = \frac{1}{2m} \times \left(\frac{h}{4\pi a_0} \right)^2 \Rightarrow \text{K.E.} = \left(\frac{h^2}{32 m \pi^2 a_0^2} \right)$$

46.(C) Use $(n + \ell)$ rule $\therefore$ Order of increasing energy is 3s, 3p, 4s, 3d.

47.(D) Number of d-electrons in $\text{Fe}^{2+} = 6$; Number of d-electrons in Fe = 6 (as in Fe^{2+} , e's are removed from 4s)
 Number of p-electron is $(1s^2 2s^2 2p^6)$ in Ne = 6 ; Number of s-electron in Mg $(1s^2 2s^2 2p^6 3s^2)$
 Number of p-electrons in Cl = 11 $(1s^2 2s^2 2p^6 3s^2 3p^5)$

48.(A) Only one orbital, 3p has following set of quantum numbers, $n = 3$, $l = 1$ and $m_l = 0$.

49.(B) The electron is more tightly bound in the smallest allowed orbit.

50.(A) Orbital angular momentum $(m) = \sqrt{\ell(\ell + 1)} \frac{h}{2\pi}$

For p-electrons ; $\ell = 1$ Thus, $m = \sqrt{\ell(\ell + 1)} \frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi} = \frac{h}{\sqrt{2}\pi}$

51.(A) Correct order of filling for $n = 6$ in outermost will be : (Aufbau principle) : 6s 4f 5d 6p

52.(B) In an atom, for any value of n , the values of $l = 0$ to $(n - 1)$. For a given value of l , the values of $m_l = -l$ to $+l$ and the value of $s = 1/2$ or $-1/2$.

53.(C) From Heisenberg uncertainty principle

$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi} \text{ or } m\Delta v \times \Delta x \geq \frac{h}{4\pi}$$

$$\text{or } (m\Delta v)^2 \geq \frac{h^2}{4\pi^2} \quad (\because \Delta x = \Delta p)$$

$$\text{or } \Delta v \geq \frac{1}{2m} \sqrt{\frac{h}{\pi}}$$

54.(C) Uncertainty in momentum $(m\Delta v) = 1 \times 10^{-18} \text{ g cm s}^{-1}$

$$\text{Uncertainty in velocity, } (\Delta v) = \frac{1 \times 10^{-18}}{9 \times 10^{-28}} = 1.1 \times 10^9 \text{ cm s}^{-1}$$

55.(B) (i) represents an electron in 3s orbital.

(ii) is not possible as value of l varies from 0 to $(n-1)$..

(iii) represents an electron in 4f orbital.

(iv) is not possible as value of m varies from $-l$ to $+l$.

(v) is not possible as value of m varies from $-l$ to $+l$, it can never be greater than l .

56.(A) $E_n = -13.6 \left(\frac{Z^2}{n^2} \right)$; Hence 1st excited state is $n = 2$: $\text{KE} = -E_n = -(-3.4) \text{ eV} = 3.4 \text{ eV}$

57.(A) $n = 3$, $l = 2$, $m = +2$ (+2, +1, 0, -1, -2). It symbolizes one of the five d-orbitals

$$58.(C) \quad \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-27} \text{ erg sec}}{1g \times 10^4 \text{ cm/s}} = 6.63 \times 10^{-31} \text{ cm} = 6.63 \times 10^{-23} \text{ m}$$

59.(A) Since both CO and CN⁻ have 14 electrons, therefore these are iso-electronic (i.e. having same number of electrons).

60.(A) The number of electrons in O²⁻, N³⁻, F⁻ and Na⁺ is 10 each, but number of electrons in Mg⁺ is 11.

$$61.(1) \quad \begin{array}{ccc} & H^1 & H^2 & H^3 \\ p & 1 & 1 & 1 = 3 \\ n & 0 & 1 & 2 = 3 \\ \therefore & \frac{3}{3} = 1 \end{array}$$

$$62.(20) \quad 100 \times 8000 = W \times 4000 \Rightarrow W = 200$$

$$63.(32) \quad \text{Total number of } e^- = 7 + 8 \times 3 + 1 = 32$$

$$64.(1) \quad \frac{hc}{\lambda} = 1 + W; \frac{3hc}{\lambda} = 5 + W \Rightarrow W = 1 \text{ eV}$$

$$65.(27) \quad kZ^2 = 36,36 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = 27 \text{ eV}$$

$$66.(5) \quad \Delta x_{\min} = \frac{h}{4\pi m \Delta V} = \frac{6.626 \times 10^{-34}}{4\pi \times 10^{-6} \times \frac{3.313}{\pi} \times 10^{-3}} = 5 \times 10^{-26} \text{ m}$$

$$67.(1) \quad \lambda = \frac{h}{\sqrt{3mkT}}, \lambda \propto m^{-1/2} T^{-1/2}$$

$$68.(45) \quad n = 12 \text{ to } n = 3; \Delta n = 9; 9 \times \frac{10}{2} = 45$$

$$69.(20) \quad 3.6 \times 1.6 \times 10^{-13} = \frac{(9 \times 10^9)(Z)(2)(1.6 \times 10^{-19})^2}{1.6 \times 10^{-14}} \\ Z = 20$$

$$70.(12) \quad \text{Intensity of each beam} = 1.2 \times 10^{-3}$$

$$\text{Wattage of each beam} = 1.2 \times 10^{-3} \times 10^{-4} = 1.2 \times 10^{-7} \text{ W}$$

$$E_1 = \frac{12400}{4144} \text{ eV}, E_2 = \frac{12400}{4972} = 2.5 \text{ eV}$$

$$E_3 = \frac{12400}{6216} = 2 \text{ eV};$$

Only E₁ and E₂ can eject photoelectrons.

$$N_1 = \frac{1.2 \times 10^{-7}}{2.5 \times 1.6 \times 10^{-19}}; N_2 = \frac{1.2 \times 10^{-7}}{3.0 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow N_1 + N_2 = 0.55 \times 10^{12} \text{ in 1 sec}$$

$$\therefore \text{in 2 sec.} = 1.1 \times 10^{12}$$

$$71.(25) \quad r_n = \frac{n^2 h^2}{4k\pi^2 \times 3e^2 \times 208m_e}$$

72.(80) $n_2 = 3, n_1 = 1$

$$\frac{1}{\lambda} = R_H \times 3^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$$

$$\Rightarrow \lambda = \frac{4}{5R_H} = 8 \times 10^{-8} \text{ m} = 80 \text{ nm}$$

73.(10) No. of photoelectrons emitted = $\frac{90}{3} \times \frac{4000 \times 10^{-10}}{6.4 \times 10^{-34} \times 3 \times 10^8} = \frac{1}{1.6} \times 10^{20}$

$$\therefore \text{Magnitude of charge passing per second} = \frac{1}{1.6} \times 10^{20} \times 1.6 \times 10^{-19} = 10 \text{ A}$$

74.(97)

75.(4) $E_{\text{incident}} = \frac{1240}{300} = 4.1 \text{ eV}$

So Li, Na, K, Mg will show.

Periodic Properties of Elements

1.(C)

2.(B) Pb has most stable +2 state among the given (inert pair effect)

3.(C) $O^{2-} > F^- > Na^+ > Mg^{+2}$

4.(D)

5.(A) Shielding effect : $s > p > d > f$

6.(A) 1st ionization enthalpies : $Na < Mg > Al < Si$

7.(C)

8.(D) $I^- > I > I^+$ (anion are larger than parent ; cations are smaller)

9.(C)

10.(A) $\left. \begin{array}{l} \text{I is Neon} \\ \text{II is oxygen} \\ \text{III is sodium} \\ \text{IV is fluorine} \end{array} \right\} \text{Electron Affinity}$
 $IV > II > III > I$

11.(BC) Sulphur and Phosphorus can show covalency greater than 4. Ex : SF_6 , PCl_5 etc.

12.(AC) Group 1 and 2 elements impart characteristic flame colour.

13.(AD) Sulphur has maximum electron affinity in its group. Chlorine has maximum electron affinity among halogens.

14.(ACD)

15.(BC)

16.(BC) (Actually 1st ionization enthalpy follows $B < C < N > O$) and 1st electron gain enthalpy $Cl > F > Br > I$)

17.(AD)

18.(AC)

19.(AC)

20.(B)

21.(A)

22.(C)

23.(B)

24.(A) Alkali metal oxides are more basic $\therefore K_2O > Na_2O > MgO > Al_2O_3$

25.(D) $O^{2-} > F^- > Na^+ > Mg^{+2} > Al^{+3}$

26.(B)

27.(C) Denser the cation, more will be the polarizing power.

28.(B) Rb^+ is least solvated

29.(B)

30.(D)

31.(C)

32.(C)

33.(C)

34.(C)

35.(D)

36.(D)

37.(B)

38.(B)

39.(B)

40.(ABCD)

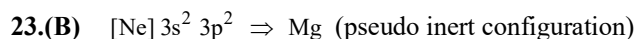
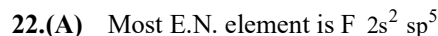
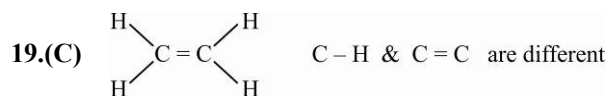
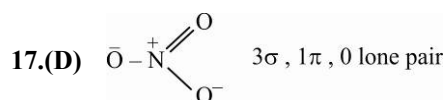
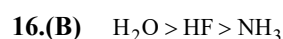
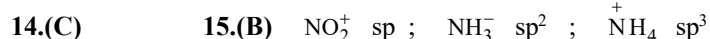
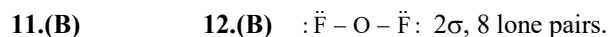
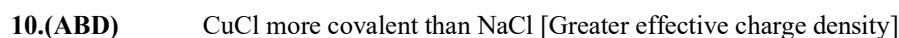
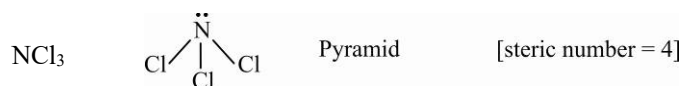
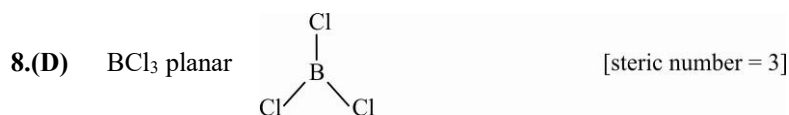
- 41.(D)** Electron gain enthalpy becomes less negative from top to bottom in a group while it becomes more negative from left to right within a period.
- 42.(A)** As positive charge on the cation increases, effective nuclear charge increases. Thus atomic size decreases.
- 43.(A)** $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$; $\Delta H = 5.1 \text{ eV}$; $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$; $\Delta H = -5.1 \text{ eV}$.
- 44.(A)** In going left to right across a period in the periodic table, the basicity (i.e. proton affinity) decreases as the electronegativity of the atom possessing the lone pair of electrons increases. Hence basicity of NH_2^- is higher than F^- . In moving down in a group, as the atomic mass increases, basicity decreases. Hence F^- is more basic than I^- and HO^- is more basic than HS^- . Hence among the given ionic species, NH_2^- has maximum proton affinity.
- 45.(A)**
- | | | | | |
|-------------------------------------|-------|---------|---------|-------|
| X – X bond | F – F | Cl – Cl | Br – Br | I – I |
| Bond dissociation energy (kcal/mol) | 38 | 57 | 45.5 | 35.6 |
- The lower value of bond dissociation energy of fluorine is due to the high inter-electronic repulsion between non-bonding electrons in the 2p-orbitals of fluorine. As a result F – F bonds is weaker in comparison to Cl – Cl and Br – Br bonds.
- 46.(B)** Amongst isoelectronic ions, ionic radii of anions is more than that of cations. Further size of the anion increases with in –ve charge and size of cation decreases with increases in +ve charge. Hence correct order is $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.
- 47.(D)** Atomic volume is the volume occupied by one gram of an element. Within a period from left to right, atomic volume first decreases and then increases.
- 48.(2)** $\text{Ti}^{2+}, \text{V}^{3+}$
- 49.(3)** 83, 54, 34
- 50.(9)** $\text{Fe}^{+3}, \text{Mn}^{2+}, \text{Co}^{4+}, \text{Co}^{3+}, \text{Cr}^+, \text{Fe}^{2+}, \text{Mn}^{3+}, \text{Cr}^{3+}, \text{V}^{3+}$
- 51.(4)** Sc, Pd, Os, Zr
- 52.(3)** $\text{Se}^{2-}, \text{N}^{3-}, \text{P}^{3-}$
- 53.(6)** Li, Na, Mg, Rb, Sr, Ca
- 54.(1)** Pt only
- 55.(4)**
- 56.(9)** $x = 4, y = 13$
- 57.(9)** $a = 0, b = 5, c = 14$
- 58.(0)**
- 59.(5)** $x = 2(\text{NO}, \text{N}_2\text{O}); y = 3(\text{MgO}, \text{SrO}, \text{K}_2\text{O})$
- 60.(4)** $n = 3, \ell = 1$
- 61.(48)** 7s, 7p, 6d, 5f
- 62.(32)**
- No. of orbitals = $1 + 3 + 5 + 7 = 16$.
- $16 \times 3 = 48$

Chemical Bonding - 1 & 2

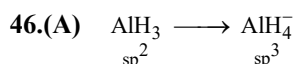
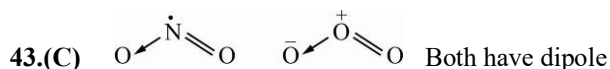
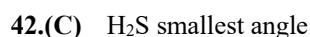
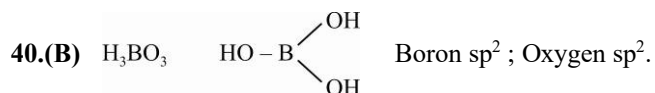
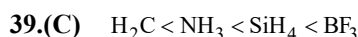
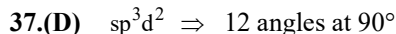
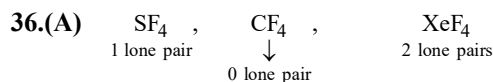
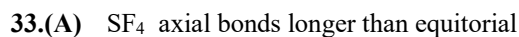
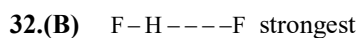
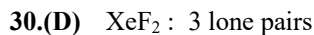
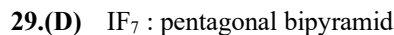
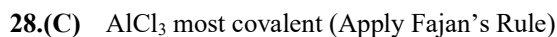
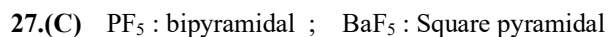
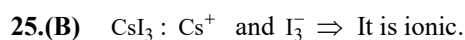
- 1.(C)** Apply Fajan's rule
- 2.(AB)**
- 3.(D)** Te^{2-} largest size, loosely packed e^-
- 4.(D)** BF_3 sp^2 , planar ; NF_3 sp^3 , pyramidal
- 5.(ABCD)**
- 6.(C)**
- $\begin{array}{c} \text{:}\ddot{\text{O}}-\text{H} \\ | \\ \text{CH}_3-\text{C}=\text{CH}_2 \end{array}$

$9\sigma, 1\pi \text{ bond}, 2 \text{ lone-pairs}$
- 7.(D)**
- $\begin{array}{c} \text{NC} \quad \quad \text{CN} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{NC} \quad \quad \text{CN} \end{array}$

$9\sigma, 9\pi$



Both bonds if cleaved one by one can never be of same strength. Successive H^+ donating ability of acids $\downarrow$ successively.



47.(B) Fe^{+3} is more polarizing than Fe^{+2} $\therefore$ FeCl_3 more covalent

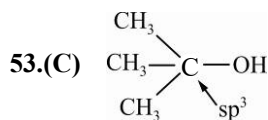
48.(A) H_2 has high vapour pressure because of intra-H-bonding

49.(B) HCl has lowest boiling point

50.(C) MX_3 ; zero dipole $\therefore$ sp^2

51.(D) $\text{Be}^{+2} < \text{Mg}^{+2} < \text{Li}^+ < \text{Na}^+$

52.(D)



54.(B)

55.(C) NF_3 and H_3O^+ iso-structural; NO_3^- and BF_3 iso-structural

56.(C) Unpaired e^- in KO_2 O_2^- (check MOT)

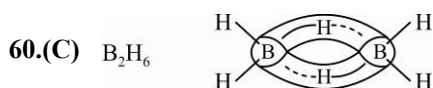
57.(B) NH_3 sp^3 ; $[\text{PtCl}_4]^{2-}$ dsp^2 ; PCl_5 sp^3d ; BCl_3 sp^2

58.(A) $\text{H}_3\text{N} \rightarrow \text{BF}_3$

sp^3 sp^3

tetrahedral tetrahedral

59.(C) SF_2 2BP 2 ℓ p sp^3 ; SF_4 4BP 1 ℓ p sp^3d ; SF_6 6BP 0 ℓ p sp^3d^2



61.(D) SO_4^{2-} sp^3 ClO_4^- sp^3 } tetrahedral

62.(A)

63.(D)

64.(B) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$
1.5 2 2.5

65.(AB) C_2 and N_2 diamagnetic

66.(C)

67.(B) Read the option (B) as $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$ (Li_2^{2-} has more electron in anti-bonding orbitals)
Bond order 0.5 0.5 1

68.(A)

69.(A)

70.(B) $\text{NO} \longrightarrow \text{NO}^+$
Para Diamagnetic

71.(C)

72.(B) Bond length $\propto \frac{1}{\text{Bond order}}$

73.(D) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$

74.(AB) Both 3

75.(AD) Both sp

76.(AB) Both $14e^-$

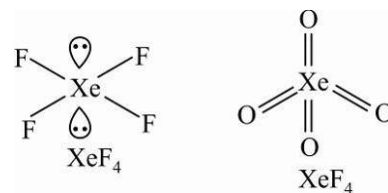
77.(CD) Both are bent

78.(CD) Check resonance

79.(AD) Check MOT

80.(CD) Both $17e^-$ species

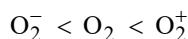
81.(C) In diamond and silicon carbide, central atom is sp^3 hybridised and hence, both are isostructural. NH_3 and PH_3 , both are pyramidal and central atom in both cases is sp^3 hybridised. SiCl_4 and PCl_4^+ , both are tetrahedral and central atom in both cases is sp^3 hybridised. In XeF_4 , Xe is sp^3d^2 hybridised and structure is square planar while in XeO_5 , Xe is sp^3 hybridised and structure is tetrahedral.



82.(B)

Species	Hybridisation	Shape	No. of e ⁻ s
SO ₃ ²⁻	sp ³	Pyramidal	42
ClO ₃ ⁻	sp ³	Pyramidal	42
CO ₃ ²⁻	sp ²	Triangular planar	32
NO ₃ ⁻	sp ²	Triangular planar	32

83.(D)



Boiling point 1.5 2.0 2.5

84.(A)

	NO ₃ ⁻	NO ₂	NO ₂ ⁻	NO ₂ ⁺
Hybridisation	sp ²	sp ²	sp ²	sp
Bond angle	120°	134°	115°	180°

85.(A) Boron hydrides are electron deficient compounds.

86.(A) HCl is polar ($\mu \neq 0$) and He is non-polar ($\mu = 0$) gives dipole-induced dipole interaction.

87.(C) M.O. configuration of O₂ $\Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1$

$$\Rightarrow \text{Paramagnetic and B.O.} = \frac{10-6}{2} = 2$$

$$O_2^+ \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \Rightarrow \text{Paramagnetic and Bond order} = \frac{10-5}{2} = 2.5$$

$$C_2 \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \Rightarrow \text{Diamagnetic and Bond order} = \frac{8-4}{2} = 2$$

$$C_2^+ \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^1 \Rightarrow \text{Paramagnetic and Bond order} = \frac{7-4}{2} = 1.5$$

$$NO \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \Rightarrow \text{Paramagnetic and Bond order} = \frac{10-5}{2} = 2.5$$

$$NO^+ \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \Rightarrow \text{Diamagnetic and Bond order} = \frac{10-4}{2} = 3$$

$$N_2 \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \Rightarrow \text{Paramagnetic and Bond order} = \frac{10-4}{2} = 3$$

$$N_2^+ \Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^1 \Rightarrow \text{Paramagnetic and Bond order} = \frac{9-4}{2} = 2.5$$

88.(D)



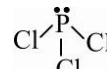
2 B.P. and 1 L.P.



3 B.P. and 0 L.P.



2 B.P. and 2 L.P.



3 B.P. and 1 L.P.

89.(D) Diatomic Species :



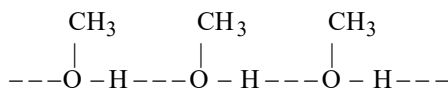
Bond order : 2.5 1.5 3.0 0.5 Thus increasing order : He₂⁺ < O₂⁻ < NO < C₂²⁻

90.(A) Ions :

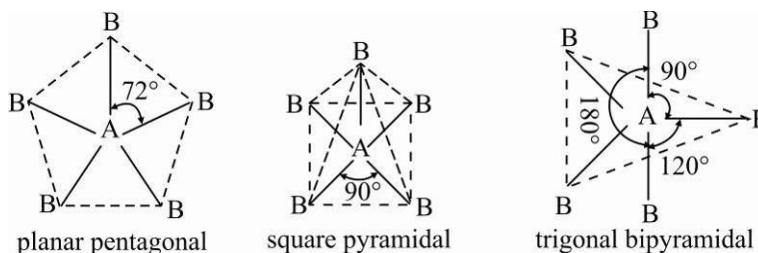


Hybridisation : sp² sp² sp³ sp³ sp

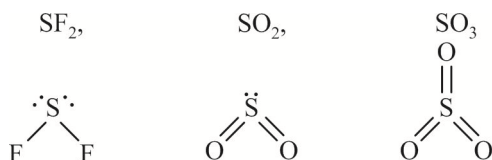
- 93.(D)** Methanol can undergo intermolecular association through H-bonding as the $-OH$ group in alcohols is highly polarized.



94.(D) For AB_5 molecules, there are three possible geometries i.e. planar pentagonal, square pyramidal and trigonal bipyramidal.



102.(3) Planar species are:



- 104.(3)** $(\text{HPO}_3)_n$ has n P—O—P linkages.

105.(3) $\text{H}_4\text{P}_2\text{O}_5, \text{H}_3\text{PO}_2, \text{H}_3\text{PO}_3$

106.(4) $\text{XeF}_2, \text{XeF}_4, \text{XeO}_6^{4-}, \text{XeCl}_4$

107.(17) $\text{P} - \text{O} - \text{P} \rightarrow 6; \text{B} - \text{O} - \text{B} \rightarrow 5; \text{Si} - \text{O} - \text{Si} \rightarrow 3; \text{S} - \text{O} - \text{S} \rightarrow 3$

108.(4) $\text{CCl}_4, \text{SiF}_4, \text{BF}_4^-, \text{SiO}_4^{4-}$

109.(1) (b) is only correct.

110.(5) Any existing species has $N_b < N_a$ electron.

111.(4) $\text{C}_2, \text{O}_2, \text{O}_2^+, \text{NO}$

112.(2) $\text{B}_3\text{N}_3\text{H}_6, \text{C}_3\text{O}_2$

113.(4)

	HOMO	Symmetry
B_2	$\pi 2p_x = \pi 2p_y$	u
N_2	$\sigma 2p_z$	g
C_2	$\pi 2p_x = \pi 2p_y$	u
N_2^{2+}	$\pi 2p_x = \pi 2p_y$	u
O_2	$\pi^* 2p_x = \pi^* 2p_y$	g
O_2^{2-}	$\pi^* 2p_x = \pi^* 2p_y$	g
N_2^{2-}	$\pi^* 2p_x = \pi^* 2p_y$	g

114.(4) $\text{NO}, \text{ClO}_2, \text{OF}, \text{B}_2$

115.(3) $\text{N}_2^+ = (13) / \text{Bond order } \text{N}_2^+ = \frac{9-4}{2} = 2.5$

[N_2^+ has one unpaired electron in $\sigma_{(2p)}$ (BMO)]

$\text{O}_2^- = (17) / \text{Bond order } \text{O}_2^- = \frac{10-7}{2} = 1.5$

[O_2^- has one unpaired electron in π^*]

$\text{OF} (17) = \text{Bond order (OF)} = \frac{10-7}{2} = 1.5$

[OF has one unpaired electron in π^*]

States of Matter

1.(B) Refer theory 2.(C) $V = \left(\frac{nR}{P}\right)T$; Slope $\propto \frac{1}{P}$ $\therefore P_1 < P_2 < P_3 < P_4$

3.(C) Refer theory from module on intermolecular forces.

4.(A) $V = C \Rightarrow P \propto T$ $\therefore T \uparrow, P \uparrow$

5.(D) $\uparrow T_C$ implies ease of liquefaction

Hence $O_2 > N_2 > H_2 > He$ Increasing order of liquefaction
 $154.3 \quad 126 \quad 33.2 \quad 5.3$

6.(B) $F = \eta A \frac{dv}{dx}$ (η : Viscosity coefficient) Observe that its unit in SI system will be $Ns m^{-2}$.

7.(A) $PV = C$ at lowest temperature for ideal gases

8.(C) Viscosity $\uparrow$ with $\uparrow$ in intermolecular forces

9.(B) Refer theory

10.(BC) By kinetic theory of gases

11.(BC) Because at high pressure and low perpendicular interaction between the particles increases.

12.(BD) $V \cdot P \propto \text{surface area} \propto \text{temperature}$

13.(A) Refer theory

14.(B) (R) is not a correct explanation of (A) but both statements are correct.

15.(C) At high altitude atmospheric pressure reduces

16.(A) Refer theory on liquefaction of gases on Andrew isotherms.

17.(D) Liquids tends to have minimum surface area in order to have minimum energy.

18.(B) At law $P \Rightarrow V_m - b \approx V_m$

$$\left(P + \frac{a}{V_m^2}\right)(V_m) = RT \Rightarrow Z = \frac{PV_m}{RT} = 1 - \frac{a}{RTV}$$

19.(C) $C^* = \sqrt{\frac{2RT}{M_0}}$; $\bar{C} = \sqrt{\frac{8RT}{\pi M_0}}$; $C = \sqrt{\frac{3RT}{M_0}}$

20.(C) At $\uparrow P \Rightarrow \left(P + \frac{a}{V_m^2}\right) \approx P$ although V_m is small

$$P(V_m - b) = RT \Rightarrow Z = \frac{PV_m}{RT} = 1 + \frac{Pb}{RT}$$

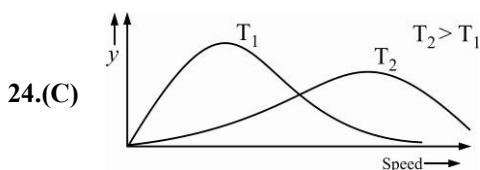
21.(D) $T_C \propto \left(\frac{a}{b}\right)$

$\uparrow T_C$ means ease of liquefaction Cl_2 is larger in size as compared to C_2H_6 $\therefore \frac{a}{(Cl_2)} > \frac{a}{C_2H_6}$ also $\frac{b}{(Cl_2)} < \frac{b}{(C_2H_6)}$

22.(D) By kinetic theory of gases.

23.(D) $\downarrow$
 $\frac{P}{\text{Vapour Pressure}} \times V = n \times R \times 300$

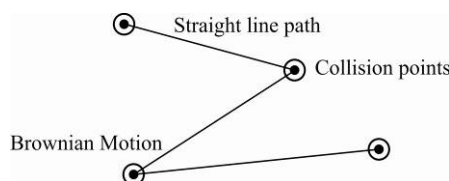
$$\left(\frac{26.7}{760}\right) \times 1 = n \times 0.0821 \times 300 \Rightarrow n = 1.42 \times 10^{-3}$$



26.(C) $b = 4 \times [\text{Na} \times \text{Volume of one particle}]$

25.(C) K.E. of an atom $= \frac{3}{2} K_B T \Rightarrow \text{K.E.} \propto T$

27.(C)

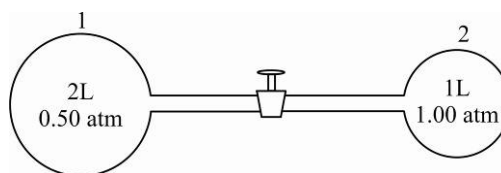


28.(D) Boyle's law, Avogadro's law, Charles's law are based on ideal gases which is explained by kinetic theory of gases.

29.(C) $PV = nRT \Rightarrow \frac{P}{RT} = \left(\frac{n}{V}\right)$

30.(B) $\frac{V}{T} = \left(\frac{nR}{P}\right) \Rightarrow P \uparrow \frac{V}{T} \downarrow$

31.(C) Initially $2 \times 0.5 = n_1 RT \dots(\alpha)$
 $1 \times 1 = n_2 RT \Rightarrow n_1 = n_2$
 Finally Let common $P_{\text{resence}} = P$
 $P \times 2 = n_1^1 RT \dots(\beta)$



$$P \times 1 = n_2^1 RT \Rightarrow \frac{n_1^1}{n_2^1} = 2 \Rightarrow n_2^1 = \frac{n_1^1}{2}$$

Also $n_1^1 + n_2^1 = 2n_1 \Rightarrow n_1^1 + \frac{n_1^1}{2} = 2n_1 \Rightarrow n_1^1 = \left(\frac{4n_1}{3}\right) \dots(\gamma)$

Re writing equation (α) , (β) and (γ)

$$P \times 2 = \left(\frac{4n_1}{3}\right) RT \Rightarrow 2 \times 0.5 = n_1 RT$$

$$\frac{P}{0.5} = \frac{4}{3} \Rightarrow P = \frac{2}{3} = 0.67$$

32.(CD) Units of Boltzmann's constant $\text{J mol}^{-2} \text{K}^{-1}$

33.(B) It is a polar molecule, thus more attractive forces among its molecules.

34.(D) C_P for monoatomic gas mixture of same volume $= \frac{5}{2}R$ and $C_V = \frac{3}{2}R \therefore \frac{C_P}{C_V} = \frac{5}{3} = 1.67$

35.(A) We know that from ideal equation : $V \propto \frac{T}{P}$

Given $T_1 = 15 + 273 = 288$; $P_1 = 1.5 \text{ bar}$; $T_2 = 25 + 273 = 298$, $P_2 = 1 \text{ bar}$

$$V_1 \propto \frac{288}{1.5} \text{ i.e. } V_1 \propto 192 \text{ and } V_2 \propto \frac{298}{1} \Rightarrow \frac{V_2}{V_1} = \frac{298}{192} = 1.55 = 1.6$$

36.(A) The average translational K.E. (E_T) of one molecule of an ideal gas will be given by $E_T = \frac{\text{K.E.}}{N_A} = \frac{3/2RT}{N_A} = \frac{3}{2}KT$

where $\frac{R}{N_A} = \text{Boltzmann constant} \Rightarrow E_T \propto T$. So, at constant temperature K.E. of molecules remains same.

37.(D) Mass of the gas = mass of the cylinder including gas – mass of empty cylinder.
So mass of a gas can be determined by weighing the container in which it is enclosed.
Thus, the statement (D) is wrong for gases.

38.(A) Temperature at which all molecular motion ceases is called absolute zero.

39.(A) According to Charles' law which states that "the volume of the given mass of a gas increases or decreases by $\frac{1}{273}$ of its volume at 0°C for each degree rise or fall of temperature at constant pressure".

$$V_t = V_0 \left(1 + \frac{t}{273} \right) \text{ at constant P and n.}$$

40.(B) $\frac{PV}{T} = \text{constant} \Rightarrow \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow \frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$

41.(4) $3.2\text{g O}_2 = 0.1\text{ mol,}$

$0.2\text{g H}_2 = 0.1\text{mol,}$

Total $n = 0.2\text{ mol,}$

$$P = \frac{nRT}{V} = \frac{0.2 \times 0.0821 \times 273}{1.12} = 4\text{ atm}$$

42.(4) Suppose molecular masses of A and B are M_A and M_B respectively. Then their number of moles will be

$$n_A = \frac{1}{M_A}, \quad n_B = \frac{2}{M_B}$$

$$P_A = 2\text{bar}, P_A + P_B = 3\text{ bar}$$

i.e. $P_B = 1\text{ bar}$

Applying the relation $PV = nRT$

$$\therefore \frac{P_A}{P_B} = \frac{n_A}{n_B} = \frac{1/M_A}{2/M_B} = \frac{M_B}{2M_A}$$

or $\frac{M_B}{M_A} = 2 \times \frac{P_A}{P_B} = 2 \times \frac{2}{1} = 4$

or $M_B = 4M_A$

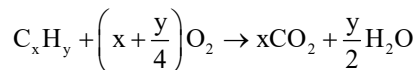
43.(44) If t is the time taken for diffusion of each gas, then by Graham's law of diffusion

$$\frac{r_x}{r_{\text{Cl}_2}} = \frac{127/t}{100/t} = \sqrt{\frac{M_{\text{Cl}_2}}{M_x}} = \sqrt{\frac{71}{M_x}}$$

or $\frac{71}{M_x} = \left(\frac{127}{100} \right)^2$

or $M_x = 44\text{ u}$

44.(56) The balanced equation for combustion



$$1 \text{ volume} \left(x + \frac{y}{4}\right) \text{ volume}$$

$$\therefore x + \frac{y}{4} = 6 \text{ (by equation)}$$

$$\text{or } 4x + y = 24 \quad \dots(1)$$

Again $x = 4$ since evolved CO_2 is 4 times that of hydrocarbon

$$\therefore 16 + y = 24 \quad \text{or } y = 8 \therefore \text{ formula of hydrocarbon } C_4H_8$$

$$45.(40) \quad n_A = \frac{2.9}{M_A}; n_{H_2} = \frac{0.184}{2} = 0.092$$

$$\frac{(PV)_A}{(PV)_{H_2}} = \frac{(nRT)_A}{(nRT)_{H_2}}$$

$$1 = \frac{2.9}{M_A} \times \frac{(95 + 273)}{(17 + 273)} \times \frac{1}{0.092}$$

$$M_A = \frac{2.9 \times 368}{290 \times 0.092} = 40 \text{ g mol}^{-1}$$

$$46.(22) \quad \frac{P_1 V_1}{P_2 V_2} = \frac{n_1 R T_1}{n_2 R T_2} = \frac{n_1 T_1}{n_2 T_2}$$

$$\text{As, } V_1 = V_2 \text{ \& } T_1 = T_2$$

$$\frac{P_1}{P_2} = \frac{n_1}{n_2}$$

$$\frac{P_{H_2}}{P_{CO_2}} = \frac{n_{H_2}}{n_{CO_2}}$$

$$\frac{P_{H_2}}{1} = \frac{44/2}{44/44} = 22 \text{ atm}$$

$$47.(1) \quad P \propto \frac{1}{V}$$

$$48.(16) \quad \frac{r_A}{r_B} = \sqrt{\frac{M_B}{M_A}}$$

$$\frac{2}{1} = \sqrt{\frac{64}{M_A}}$$

$$M_A = 16 \text{ g / mole}$$

49.(3) $PV = nRT$

$$\frac{(PV)_i}{(PV)_f} = \frac{(nRT)_i}{(nRT)_f}$$

$$\frac{10V}{8V} = \frac{\left(\frac{15}{MM}\right)}{\frac{W}{MM}}$$

$$\frac{10}{8} = \frac{15}{W}$$

$$W = 12 \text{ kg}$$

$$\text{Leaked gas} = (15 - 12) = 3 \text{ kg}$$

50.(327) $\left(\sqrt{\frac{2RT}{M}}\right)_{\text{SO}_2} = \left(\sqrt{\frac{2RT}{M}}\right)_{\text{O}_2}$

$$\left(\frac{T}{M}\right)_{\text{SO}_2} = \left(\frac{T}{M}\right)_{\text{O}_2}$$

$$\frac{T_{\text{SO}_2}}{64} = \frac{300}{32} \Rightarrow T_{\text{SO}_2} = 600 \text{ K} = 327^\circ \text{C}$$

51.(3) $\text{rms}(\mu) = \sqrt{\frac{3RT}{M}}$

$$\text{For atom, } \mu' = \left(\sqrt{\frac{3RT}{M}}\right)'$$

$$T' = 3T \text{ and } M = 3M'$$

$$\frac{\mu'}{\mu} = \frac{\left(\sqrt{\frac{3RT}{M}}\right)'}{\left(\sqrt{\frac{3RT}{M}}\right)} = \sqrt{\frac{3R \times 3T}{\frac{M'}{3M}}} = \sqrt{9} = 3$$

$$\mu' = 3\mu$$

52.(500) Let the initial number of moles in vessel be 5 at 27°C , i.e., 300K. Moles of air left at new temperature

$$= 5 - \frac{2}{5} \times 5 = 3$$

Pressure and volume remaining the same,

$$n_1RT_1 = n_2RT_2$$

$$5 \times 300 = 3 \times T_2 \Rightarrow T_2 = 500 \text{ K}$$

53.(32) $\frac{r_x}{r_y} = \frac{\frac{V_x}{t_x}}{\frac{V_y}{t_y}} = \frac{\frac{40}{20}}{\frac{80}{20}} = \frac{1}{2} \Rightarrow \sqrt{\frac{M_x}{M_y}} = \frac{r_y}{r_x} \Rightarrow \frac{M_x}{M_y} = 4$

$$M_y = \frac{128}{4} = 32$$

$$54.(400) T_c = \frac{8 P_c V_c}{3 R} = \frac{8}{3} \times \frac{40 \times 3}{0.8} = 400K$$

$$55.(3) PM = \rho RT \Rightarrow \frac{(PM)_{\text{given}}}{(PM)_{\text{STP}}} = \frac{(\rho RT)_{\text{given}}}{(\rho RT)_{\text{STP}}}$$

$$\frac{2}{1} = \frac{5.46 \times 300}{\rho \times 273} \Rightarrow \rho = 3g \text{ dm}^{-3}$$

Thermochemistry & Thermodynamics

1.(C) Closed system allows heat exchange but not mass between system and surrounding.

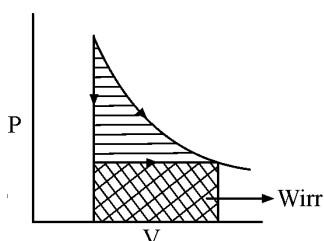
2.(C) Specific heat is constant

3.(B)

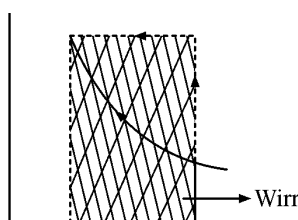
4.(C) For adiabatic process, $q = 0 \Rightarrow w = \Delta U$

Free expansion $\Rightarrow w = 0, \Delta U = 0 \Rightarrow \Delta T = 0$

5.(B)

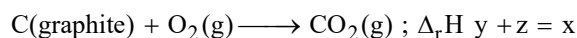
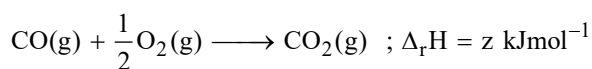
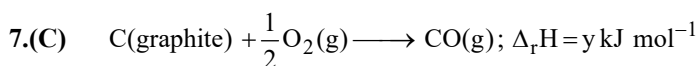


Expansion (-ve work)

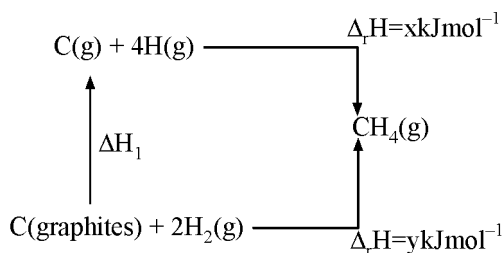


Compression (+ve work)

$$6.(C) \Delta S_{\text{system}} + \Delta S_{\text{surrounding}} \geq 0$$



8.(C) Thus $y = \Delta H_1 + x \Rightarrow x > y$ since $\Delta H_1 = +ve$ and x and y are negative.



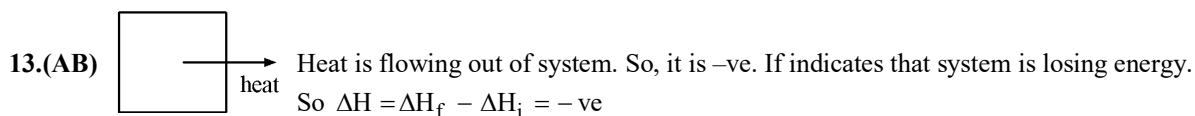
9.(C) Enthalpy of formation can be +ve or -ve

10.(A)

11.(B) For spontaneous process ΔG should be negative

12.(AD)

Chemical Rxn



14.(CD) Flow of heat from warmer body to colder body is spontaneous. In general dissipation of energy is spontaneous process.

15.(CD) $w_1 = -nR \times 300 \times \ln 10$; $w_2 = -nR \times 600 \times \ln 10$

$$\frac{w_1}{w_2} = \frac{1}{2} \Rightarrow w_2 = 2w_1 \quad \text{Also for ideal gases } \Delta U = U(T) \Rightarrow \text{for isothermal process } \Delta U = 0$$

16.(AC) $\Delta_r H = \sum \Delta H(\text{Product}) - \sum \Delta H(\text{Reactant})$

$$\text{If } \Delta_r H = -ve \Rightarrow \sum \Delta H(\text{products}) < \sum \Delta H(\text{Reactants})$$

Since energy of product is reduced it means. Some energy is evolved for the system

17.(B) 18.(A) $\Delta_C H = q_p = q_r + \Delta H_g RT = -1364.47 - \frac{1 \times 8.314 \times 298}{1000}$ $\Delta_C H = -1366.95 \text{ kJ / mol}$

19.(A) $q = 208 \text{ J}$ since system absorbs heat

$$w = -nRT \ln \frac{v_2}{v_1} = -0.04 \times 8.314 \times 310 \times \ln \left(\frac{375}{50} \right) = -207.7 \text{ J} \approx -208 \text{ J}$$

20.(C) For (A) : $\Delta G = \Delta H - T\Delta S$ For adiabatic process OR for a free expansion of a gas ; $\Delta H = 0 \therefore$

$$\frac{\Delta G_{\text{system}}}{\Delta S_{\text{system}}} = -T$$

$$\text{For (C) : } \Delta G = -nRT \ln K \Rightarrow \Delta G = \Delta H - T\Delta S$$

21.(A) $\Delta S = nR \ln \frac{v_f}{v_i} = 2 \times 8.314 \times \ln 10 = 38.3 \text{ J / K}$

22.(D) $\text{PbO}_2 + \text{Pb} \longrightarrow 2\text{PbO}$, $\Delta G^\circ < 0 \Rightarrow$ spontaneous

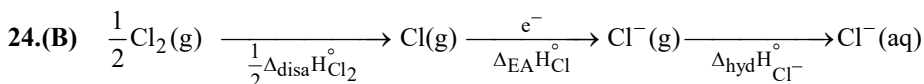
$\text{SnO}_2 + \text{Sn} \longrightarrow 2\text{SnO}$, $\Delta G^\circ > 0 \Rightarrow$ non-spontaneous

Thus O.S. for lead = +2 ; O.S for tin = +4

23.(B) $\Delta G = \Delta H - T\Delta S$

$$\text{For spontaneous } \Delta G < 0 \Rightarrow T > \frac{\Delta H}{\Delta S}$$

$$\text{But at equilibrium } \Delta G = 0 \Rightarrow \Delta H - T_e \Delta S = 0 \Rightarrow T_e = \frac{\Delta H}{\Delta S} \quad \text{Thus } T > T_e$$



$$\text{Thus energy involved in } \frac{1}{2}\text{Cl}_2(\text{g}) \longrightarrow \text{Cl}^-(\text{g}) \text{ is : } \frac{1}{2}\Delta_{\text{diss}}H_{\text{Cl}_2}^\circ + \Delta_{\text{EA}}H_{\text{Cl}}^\circ + \Delta_{\text{hyd}}H_{\text{Cl}^-}^\circ = -610 \text{ kJ / mol}$$

25.(A) $\Delta S_{\text{system}} + \Delta S_{\text{surrounding}} \geq 0$ for spontaneous process

$$\text{For isolated system } \Delta S_{\text{surrounding}} = 0 \quad \text{Thus } \Delta S_{\text{system}} \geq 0$$

26.(A) For isolated system $q = 0$

$$\text{For ideal gas } \Delta U = nC_v\Delta T = nC_v(T_f - T_i) ; \quad \text{For reversible process } \Delta U_{\text{rev}} = W_{\text{rev}}$$

$$\text{For irreversible process } \Delta U_{\text{irr}} = W_{\text{irr}} ; \quad \text{But for same expansion } |W_{\text{rev}}| > |W_{\text{irr}}|$$

$$|\Delta U_{\text{rev}}| > |\Delta U_{\text{irr}}|$$

$$|\Delta T_{\text{fer}}| > |\Delta T_{\text{irr}}| \Rightarrow (T_f)_{\text{irr}} > (T_f)_{\text{rev}}$$

27.(B) At const T and P, $\Delta S = nr \ln \frac{V_f}{V_i}$

Since work is done by system $V_f > V_i \Rightarrow \Delta S > 0$ Also $dG = dH - Tds$

For ideal gas $dU=0$ at constant T $\Rightarrow q = W = -ve$ and $dH = q \Rightarrow dG = -ve$

Thus $(ds) > 0$ and $(dG) < 0$

28.(B) $C_p - C_v = R$

C_p = Heat transfer at constant V + energy transferred to surrounding as work.

Thus $C_p - C_v = R$ = energy transferred to surrounding as mechanical work.

29.(C) Since ΔH is negative, increasing temperature equilibrium will shift on reverse direction and solubility will decrease.

30.(B) Using Gibb's Helmholtz equation : $\Delta G = \Delta H - T\Delta S$

During adsorption of a gas, entropy decreases i.e. $\Delta S < 0$

For spontaneous adsorption, ΔG should be negative, which is possible when ΔH is highly negative.

31.(A) $\Delta H = (E_a)_f - (E_a)_b = 0$.

32.(A) $\Delta_{\text{vap}}H^\circ = 40.66 \text{ kJ mol}^{-1}$

$T = 100 + 273 = 373 \text{ K}$, $\Delta E = ?$

$\Delta H = \Delta E + \Delta n_g RT \Rightarrow \Delta E = \Delta H - \Delta n_g RT$

$\text{H}_2\text{O}_{(l)} \rightleftharpoons \text{H}_2\text{O}_{(g)} \quad \Delta n_g = 1 - 0 = 1$

$\Delta E = \Delta H - RT = (40.66 \times 10^3) - (8.314 \times 373) = 37559 \text{ J/mol}$ or 37.56 kJ/mol

33.(C) For free expansion of an ideal gas under adiabatic condition $q = 0$, $\Delta T = 0$, $w = 0$.

34.(D) Gas phase reaction : $\text{PCl}_{5(g)} \rightleftharpoons \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$. It is a dissociation reaction, hence $\Delta H = +ve$. Also one mole of PCl_5 is dissociated into two moles of PCl_3 and Cl_2 in the same phase. Therefore, $\Delta S = S_{\text{product}} - S_{\text{reactant}} \Rightarrow \Delta S = +ve$.

35.(C) The criteria for spontaneity of a process in terms of ΔG is as follows :

* If ΔG is negative, the process is spontaneous.

* If ΔG is positive, the process does not occur in the forward direction. It may occur in the backward direction.

* If ΔG is zero, the system is in equilibrium.

The process does not occur.

36.(C) $\Delta G = \Delta H - T\Delta S$

$\Delta G = -ve$ for spontaneous reaction

When $\Delta S = +ve$, $\Delta H = +ve$ and $T\Delta S < \Delta H \Rightarrow \Delta G = -ve$

When randomness favours and energy factor do not favour then spontaneity depends upon the temperature.

37.(A) For spontaneous process, ΔS_{Total} i.e., $\Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$ should be positive.

- 38.(C) The entropy of a substance increases with increase in temperature. However at absolute zero the entropy of a pure crystalline substance is taken as zero, which is also called as third law of thermodynamics.
- 39.(B) Since a catalyst affect equally both forward and backward reactions, therefore it does not affect equilibrium constant of reaction.
- 40.(C) If more trans-2-pentene is added, then its concentration in right hand side will increase. But in order to maintain the constant K, concentration of cis-2-pentene will also increase. Therefore more cis-2-pentene will be formed.
- 41.(810) Since, the external pressure is greatly different from the pressure of N_2 and thus, process is irreversible.

$$\therefore W = -P_{\text{ex}} (V_2 - V_1)$$

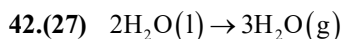
$$W = -1 \times (V_2 - V_1)$$

$$\text{Given, } V_1 = 2\text{L } V_2 = ? \quad T = 273\text{K, } P_1 = 5\text{atm, } P_2 = 1\text{atm} \therefore P_1 V_1 = P_2 V_2$$

$$\therefore V_2 = \frac{2 \times 5}{1} = 10\text{L}$$

$$\therefore W = -1 \times (10 - 2) = -8\text{L-atm} = -810\text{J}$$

$$\therefore W \times (-1) = 810\text{J}$$



$$\Delta n_g = 3 - 0 = 3$$

$$\therefore \Delta H = \Delta U + \Delta n_g RT$$

$$\Delta U = \Delta H - \Delta n_g RT$$

$$30 - 3 \times 0.0821 \times 500 \times 10^{-3}$$

$$\Delta U = 27 \text{ kcal}$$

- 43.(33) For adiabatic process

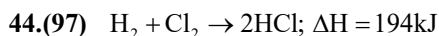
$$TV^{\gamma-1} = \text{constant}$$

For poly atomic gas

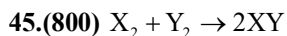
$$\gamma = \frac{4}{3} = 1.33 \quad \left(\gamma = \frac{C_p}{C_v} \right)$$

$$n = 1.33 - 1 = 0.33$$

$$\therefore n \times 100 = 33$$



$$\therefore \text{Heat of formation of HCl} = \frac{194}{2} + 97\text{kJ}$$



$$\Delta H = (\text{BE})_{x-x} + (\text{BE})_{y-y} - 2(\text{BE})_{x-y}$$

If BE of $(x-y) = a$, then BE of $(x-x) = a$

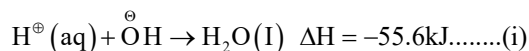
$$\text{and BE of } (y-y) = \frac{a}{2}$$

$$\Delta H_f (x-y) = -200\text{kJ}$$

$$\therefore 400 = \frac{a}{2} \text{ or } a = +800\text{kJ}$$

The bond dissociation energy of $x_2 = 800 \text{ kJ/mol}$

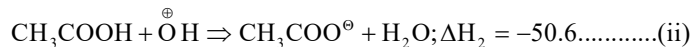
- 46.(5) The neutralization of a strong acid by a strong base is represented by



We have to calculate:



Given:



Subtract equation (i) from equation (ii), we get

$$\begin{aligned} \Delta H &= \Delta H_2 - \Delta H_1 \\ &= -50.6 - (-55.6) = 5.0 \text{ kJ mol}^{-1} \end{aligned}$$

47. (482) $\text{CH}_3\text{OH}(\text{l}) + \frac{3}{2} \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}); \Delta H = -723 \text{ kJ}$

By burning with $\frac{3}{2} \text{O}_2(\text{g})$ heat evolved = -723 kJ

$$\therefore \text{By burning with 1 mole } \text{O}_2(\text{g}) = \frac{-723 \times 2}{3} = 482 \text{ kJ}$$

- 48.(150) $\therefore 150 \text{ kJ}$ energy required for muscular work to walk a distance of one km. (Given)

$\therefore 750 \text{ kJ}$ energy required for work to walk of 5 km distance. Enthalpy of combustion is 3000 kJ mol^{-1} .

Glucose provide 30% energy of its mass.

Molecular weight of glucose = 180

Weight of glucose required for work to 5 km (750 kJ energy)

$$= \frac{180}{3000} \times 750 \times \frac{100}{30} = 150 \text{ g}$$

- 49.(300) $\frac{\Delta H_{\text{fusion}}}{T_{\text{mp}}} = \Delta S_{\text{fusion}}$

$$\frac{600}{T} = 2$$

$$T = 300 \text{ K}$$

- 50.(0) From $\Delta S_{\text{fu}} = \frac{\Delta H_{\text{fu}}}{T_{\text{f}}}$

$$T_{\text{f}} = \frac{1.435 \times 1000}{5.26} = 272.81 \approx 273 \text{ K}$$

$$\text{In } ^{\circ}\text{C} = 0$$

- 51.(0) Form $\Delta G_{\text{r}} = \Delta H_{\text{r}} - T\Delta S_{\text{r}}$

$$\Delta G_{\text{r}} = 40.63 - \frac{373.4 \times 108.8}{1000} = 0$$

- 52.(1) $\therefore \Delta G^0 = -2.303RT \log K$

$$\therefore \log K = \frac{4.606 \times 1000}{2.303 \times 2 \times 500} = 2 = \log 100$$

$$\therefore K = 1 \times 10^2$$

$$\therefore x = 1$$

53.(5) Net work done = $W_A + W_B = 25 - 20 = 5$ unit

54.(2) From $q_p = q_v + \Delta n_g RT$

$$(q_v - q_p) = \Delta n_g RT$$

$$\Rightarrow xRT = (-2)RT$$

$$x = 2$$

55. (0) From first law of thermodynamics,

$$\Delta U = Q + W$$

For isothermal process, $\Delta T = 0$

Hence, $\Delta U = 0$

Chemical & Ionic Equilibrium

1.(D) $\Delta G^\circ = -RT \ln K_{eq}$

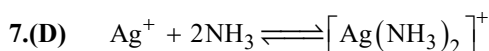
2.(C)

3.(ACD)

4.(A) On cooling reaction goes in exothermic direction.

5.(D)

6.(C)



8.(A)

9.(B) Higher bp $\Rightarrow$ lower vapor pressure

10.(ACD) $T \uparrow \Rightarrow K_{eq} \uparrow \Rightarrow$ reaction goes forward it is endothermic

11.(AD) Melting point and freezing point are same

12.(A)

13.(B) Common ion effect

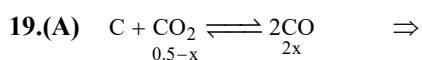
14.(A) Weak acid weak base salt solution is acidic if $K_a > K_b$; Basic if $K_b > K_a$

15.(D)

16.(C) He does to react with any of gases present Addition of inert gas at constant pressure decreases the dissociation of PCl_5 .

17.(B) $K_p = K_c (RT)^{\Delta n_g}$

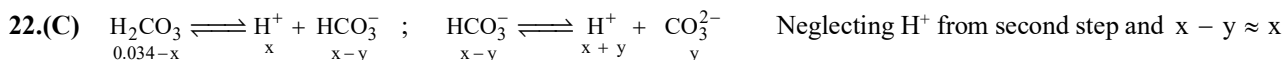
18.(D) $[H^+]$ is changed from 10^{-1} to 10^{-2} Ten times dilution is required.



$$\Rightarrow x = 0.3 \text{ atm}$$

20.(A) In III $H_2PO_4^-$ is acting like base

21.(A)



23.(B) When ppt starts $[Ag^+][Br^-] = K_{sp}(AgBr)$

24.(B) When ppt begins $[Mg^{2+}][OH^-]^2 = 10^{-11}$

25.(C) HCO_3^- is weakest. Check individual options.

26.(C) Use $pH = 7 + \frac{1}{2}(pK_a - pK_b)$

27.(A) Remove H^+

28.(A)

29.(D)

30.(C) On decreasing temperature reaction goes in exothermic direction. On increasing pressure reaction goes where number of gas particles decreases.

31.(C) NH_3 can extract proton from H_2O

32.(ABD) pH of HCl would be 6.97 approx. Use Faradays law for (D)

33.(A) HCl is a strong acid and dissociates completely into ions in aqueous solution.

34.(D) Acidic buffer is a mixture of a weak acid and its salt with a strong base. HClO₄ is strong acid.

35.(B)

Salt	K _{SP}	Solubility
Ag ₂ CrO ₄	$1.1 \times 10^{-12} = 4s^3$	$s = \sqrt[3]{\frac{K_{SP}}{4}} = 0.65 \times 10^{-4}$
AgCl	$1.8 \times 10^{-10} = s^2$	$s = \sqrt{K_{SP}} = 0.71 \times 10^{-6}$
AgBr	$5 \times 10^{-13} = s^2$	$s = \sqrt{K_{SP}} = 0.7 \times 10^{-6}$
AgI	$8.3 \times 10^{-17} = s^2$	$s = \sqrt{K_{SP}} = 0.9 \times 10^{-8}$

Solubility of Ag₂CrO₄ is highest thus, it will be precipitated at last.

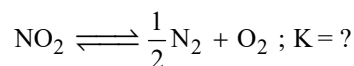
36.(C) Na₂CO₃ which is a salt of NaOH (strong base) and H₂CO₃ (weak acid) will produce a basic solution with pH greater than 7.

37.(A) BF₃ is Lewis acid (e⁻ pair acceptor).

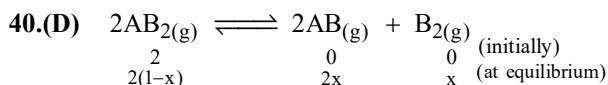
$$38.(B) \quad \text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \Rightarrow 5 = -\log K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad [\because \text{pK}_a = -\log K_a]$$

$$5 = -\log [1 \times 10^{-4}] + \log \frac{[\text{Salt}]}{[\text{Acid}]} \Rightarrow 1 = \log \frac{[\text{Salt}]}{[\text{Acid}]} \Rightarrow \frac{[\text{Salt}]}{[\text{Acid}]} = 10 = 10 : 1$$

39.(C) $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO} ; K_1$; $2\text{NO} + \text{O}_2 \rightleftharpoons 2\text{NO}_2 ; K_2$



$$K_1 = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]} ; K_2 = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]} \Rightarrow K = \frac{[\text{N}_2]^{1/2}[\text{O}_2]}{[\text{NO}_2]} = \sqrt{\frac{[\text{N}_2][\text{O}_2] \times [\text{NO}]^2[\text{O}_2]}{[\text{NO}]^2 \times [\text{NO}_2]^2}} = \sqrt{\frac{1}{K_1 K_2}}$$



Amount of moles at equilibrium = $2(1-x) + 2x + x = 2+x$

$$K_p = \frac{[\text{P}_{AB}]^2 [\text{P}_{B_2}]}{[\text{P}_{AB_2}]^2} = \frac{\left(\frac{2x}{2+x} \times P\right)^2 \times \left(\frac{x}{2+x} \times P\right)}{\left(\frac{2(1-x)}{2+x} \times P\right)^2} = \frac{\frac{4x^3}{2+x} \times P}{4(1-x)^2} \Rightarrow K_p = \frac{4x^3 \times P}{2} \times \frac{1}{4} (\because 1-x \approx 1 \& 2+x \approx 2)$$

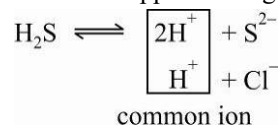
$$x = \left(\frac{8K_p}{4P}\right)^{1/3} \Rightarrow x = \left(\frac{2K_p}{P}\right)^{1/3}$$

41.(A) The cation of group II are precipitated as their sulphides.

* Solubility product of sulphide of group II radicals are very low.

* Therefore, even with low conc. of S²⁻ ions, the ionic product exceeds the value of their solubility product and the radical of group II gets precipitated.

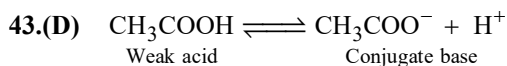
- * The low conc. of S^{2-} ions is obtained by passing H_2S gas through the solution of the salt in the presence of dil. HCl which suppresses degree of ionisation of H_2S by common ion effect.



Note: Solubility product of group IV radicals are quite high.

- * It is necessary to suppress the conc. of S^{2-} ions, otherwise radicals of group IV will also get precipitated along with group II radicals.

42.(B) CH_3COOH (weak acid), $NaOH$ (strong base). After neutralisation, the salt (CH_3COONa) formed dissociates in water giving free OH^- ions. Thus pH will be greater than 7.



As CH_3COOH is the weakest acid, so its conjugate base (CH_3COO^-) is the strongest base. H_2SO_4 , HCl , HNO_3 are strong acids, so their conjugate bases are weak.

44.(A) Vapour pressure is directly related to temperature. Greater is the temperature, greater will be the vapour pressure. So to keep it constant, temperature should be constant.

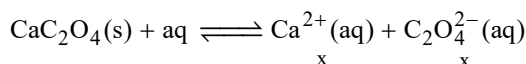
45.(C) Na_2CO_3 is salt of strong base $NaOH$ and weak acid. $NaHCO_3$ however is also having an acidic H so less basic than Na_2CO_3 .

46.(B) For precipitation : $IP \geq K_{sp}$

To initiate the precipitation : $IP = K_{sp} \Rightarrow [Mg^{2+}][OH^-]^2 = K_{sp}$

$$\Rightarrow [OH^-] = \sqrt{\frac{K_{sp}}{[Mg^{2+}]}} = \sqrt{\frac{1 \times 10^{-11}}{0.1}} = 10^{-5} M \Rightarrow pOH = 5 \Rightarrow pH = 9$$

47.(A) First find solubility in mol/L. (x mol/L)



$$\Rightarrow K_{sp} = [Ca^{2+}][C_2O_4^{2-}] = x^2 \Rightarrow x = 5 \times 10^{-5} \text{ mol/L}$$

$$\text{Solubility in g/L} = (5 \times 10^{-5}) \times 128 = 0.0064 \text{ g}$$

48.(B) mmol of $CH_3COOH = 10$; mmol of $CH_3COONa = 10$

$$pH = pK_a + \log \frac{[CH_3COONa]}{[CH_3COOH]} = pK_a + \log \frac{10/V}{10/V} = pK_a = 4.76$$

49.(B) $pH = pK_a + \log \frac{[CH_3COONa]}{[CH_3COOH]}$

$$\text{Now } \frac{[CH_3COONa]}{[CH_3COOH]} = 10 \Rightarrow \text{pH increase by 1.}$$

50.(B) $K_a = C_1 \alpha_1^2 = C_2 \alpha_2^2$; $\alpha_2 = 2\alpha_1 \Rightarrow C_1 \left[\frac{\alpha_1}{\alpha_2} \right]^2 = C_2 \Rightarrow C_2 = 0.1 \times \left[\frac{1}{2} \right]^2 = \frac{0.1}{4}$

Since concentration is decreased to $1/4m$, the volume is increased to 4 times which means volume is 200 mL. So 900 mL H_2O is added.

51.(16) $\Delta H = \Delta E + \Delta nRT$

or $-\Delta nRT = \Delta E - \Delta H = 1200$

or $\Delta n = \frac{-1200}{2 \times 300} = -2$

$\frac{K_p}{K_c} = (RT)^{\Delta n} = (0.0821 \times 300)^{-2} = (24.63)^{-2}$

or $\frac{K_p}{K_c} = \frac{1}{(24.63)^2} = 1.648 \times 10^{-3}$

or $\frac{K_p}{K_c} \times 10^4 = 16.48 \approx 16$



(moles before reaction) 2 0.5 0

after reaction $\rightarrow$ 2 - 1 = 1 0 1

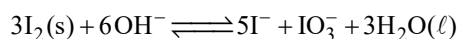
$\Delta n = |2.5 - 2| = 0.5$

Change in pressure

$\Delta P = \frac{\Delta nRT}{V} = 0.5 \times \frac{1}{12} \times \frac{300}{6.25} \text{ atm}$

$= 2$

53.(8) Balanced equation is –



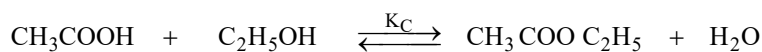
$\Delta G^\circ = 5 \times (-50) + (-123.5) + 3 \times (-233) \times -0 - 6 \times (-150) = -172.5 \text{ KJ/mol}$

Now, $\Delta G^\circ = -RT \ln k$

$\Rightarrow -172.5 = -\frac{25}{3} \times 300 \times 2.3 \times 10^{-3} \log K \quad \Rightarrow \quad \log K = 30$

$\Rightarrow 10^{30} = \frac{[\text{I}^-]^5 [\text{IO}_3^-]}{[\text{OH}^-]^6} = \frac{10^{-5} \times 10^{-1}}{[\text{OH}^-]^6}$

$\Rightarrow [\text{OH}^-] = 10^{-6} \quad \Rightarrow \quad \text{pOH} = 6 \quad \Rightarrow \quad \text{pH} = 8$



54.(2) Initially: 0.1 mol 0.1 mol 0 0

At equilibrium: (0.1 - x) mol (0.1 - x) mol x x

In 10 ml solution at equilibrium moles of CH_3COOH present = 0.1 (0.1 - x) $\equiv$ 0.1 (0.1 - x) moles of NaOH

$\therefore 0.1 (0.1 - x) = 80 \times 10^{-4} = 8 \times 10^{-3}$

or $0.01 - 0.1 x = 0.008 \quad \text{or} \quad x = 0.02$

$K_C = \frac{x^2}{(0.1 - x)^2} = \frac{(0.02)^2}{(0.08)^2} = \frac{1}{4^2} = \frac{1}{16}$

$= 32 K_C = 32 \times \frac{1}{16} = 2$

$$55.(36) \log K_p = \frac{-\Delta_r H^0}{2.303 RT} + C$$

$$\text{Slope of the plot is, } \frac{-\Delta_r H^0}{2.303 R} = 5 \times 10^3$$

$$\Delta_r H^0 = -5 \times 2.303 \times 8.314 \text{ kJ/mol}$$

$$\Delta_r H^0 = -95.736 \text{ kJ/mol}$$

$$\Delta_r S^0 = 250 - 250 - 100 = -0.1 \text{ kJ/mol}$$

$$\Delta_r G^0 = -95.736 + 298 (0.1) \text{ kJ/mol}$$

$$\Delta_r G^0 = -65.936 \text{ kJ/mol}$$

$$\log K_p = \frac{65.936 \times 10^3}{8.314 \times 298 \times 2.303} = 11.556$$

$$\text{or } K_p = 3.597 \times 10^{11} = 35.97 \times 10^{10} \approx 36 \times 10^{10}$$

$$56.(10) K_{a1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3]}, K_{a2} = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]}$$

$$K_{a1} \times K_{a2} = \frac{[H^+]^2 [CO_3^{2-}]}{[H_2CO_3]} = 10^{-18}$$

$$\therefore \frac{10^{-10} \times [CO_3^{2-}]}{10^{-2}} = 10^{-18}$$

$$[CO_3^{2-}] = 10^{-10}$$

$$\therefore -\log[CO_3^{2-}] = 10$$

$$57.(900) \because \alpha \text{ is negligible w.r.t. } 1$$

$$K_a = C_1 \alpha_1^2 = C_2 \alpha_2^2$$

$$\therefore C_2 = C_1 \left(\frac{\alpha_1}{\alpha_2} \right)^2$$

$$= 0.2 \times \frac{1}{4} = 0.05$$

$$C_1 V_1 = C_2 V_2$$

$$300 \times 0.2 = 0.05 \times V_2$$

$$V_2 = 1200 \text{ ml}$$

$$\text{Volume of } H_2O \text{ added} = 1200 - 300$$

$$= 900 \text{ ml}$$

$$58.(4) [OH^-] = [OH^-]_{NaOH} + [OH^-]_{Mg(OH)_2}$$

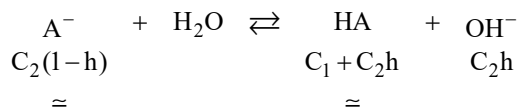
$$= 0.1 + 2s \approx 0.1 \text{ M } (2s \ll 0.1)$$

$$K_{sp}[Mg(OH)_2] = [Mg^{2+}](0.1)^2$$

$$Mg^{2+} = 4 \times 10^{-10} \text{ M}$$

59.(1) For acid buffer,

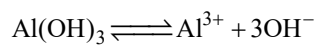
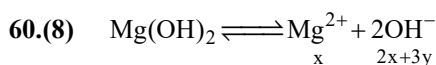
$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]} = 5 + \log \frac{0.1}{0.01} = 6$$



$$\therefore \frac{\text{C}_2h \times \text{C}_1}{\text{C}_2} = \text{K}_h = \frac{\text{K}_w}{\text{K}_a}$$

$$\therefore h = \frac{\text{K}_w}{\text{K}_a \text{C}_1} = \frac{10^{-14}}{10^{-5} \times 10^{-2}} = 10^{-7}$$

$$\therefore h \times 10^7 = 1$$



Now $x \gg y$ because K_{sp} of $\text{Mg}(\text{OH})_2 \gg \text{K}_{\text{sp}}$ of $\text{Al}(\text{OH})_3$

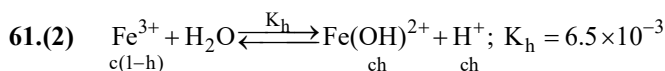
$$\Rightarrow x \times (2x)^2 = 4 \times 10^{-12} \Rightarrow x = 10^{-4} \quad \dots(\text{i})$$

$$\text{And } y \times (2x)^3 = 1 \times 10^{-33}$$

$$\Rightarrow y = \frac{10^{-33}}{8x^3} = \frac{10^{-33}}{8 \times 10^{-12}} = \frac{10^{-21}}{8} \quad \dots(\text{ii})$$

$$\frac{x}{y} = \frac{10^{-4}}{10^{-21}} \times 8 = 8 \times 10^{17}$$

$$\text{So answer} \Rightarrow 8 \times 10^{+17} \times 10^{-17} = 8$$

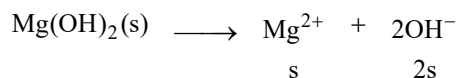
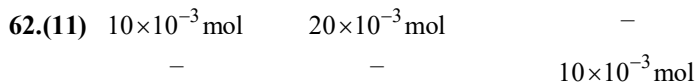


$$\text{K}_h = 6.5 \times 10^{-3} = \frac{\text{ch}^2}{(1-h)} = \frac{\text{c} \times (0.5)^2}{(0.5)} = \text{c} \times 0.5$$

$$\text{or } \text{c} = 13 \times 10^{-3} (\text{M})$$

$$[\text{H}^+] = \text{ch} = 13 \times 10^{-3} \times \frac{1}{2} = 6.5 \times 10^{-3}$$

$$\text{or } \text{pH} = 3 - \log 6.5 \times 2.187 \approx 2$$



Where s is the solubility of $\text{Mg}(\text{OH})_2$ in the resulting solution

$$\therefore 4s^3 = 12 \times 10^{-12}$$

$$\therefore s = (3)^{1/3} \times (10^{-12})^{1/3} = 1.44 \times 10^{-4} \text{ (M)}$$

$$\therefore [\text{OH}^-] = 2s = 2.88 \times 10^{-4} \text{ (M)}$$

$$\text{or } \text{pOH} = 4 - \log 2.88 = 4 - 0.459 = 2.541$$

$$\therefore \text{pH} = 14 - \text{pOH} = 14 - 2.541 = 11.459$$

63.(93) In the resulting solution, $[\text{Ca}^{2+}] = \frac{10^{-5}}{2} \text{ (M)}$

In order to observe the precipitation of CaCO_3 required $[\text{Ca}^{2+}] = 7 \times 10^{-5} \text{ (M)}$

$$\therefore \frac{7 \times 10^{-5}}{10^{-5/2}} = \frac{200}{V} \text{ where } V \text{ is the final volume of solution in ml}$$

$$\text{or } \frac{200}{V} = 14 \quad V = \frac{200}{14} \text{ ml}$$

$$\therefore \% \text{ reduction in volume} = \frac{(200 - 200/14)}{200} \times 100$$

$$= \left(1 - \frac{1}{14}\right) \times 100 = \frac{13}{14} \times 100 = 92.856 \approx 93\%$$

64.(15) For the precipitation of ZnS , required $[\text{S}^{2-}]$

$$= \frac{10^{-21}}{10^{-2}} = 10^{-19} \text{ (M)}$$

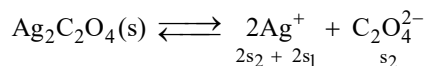
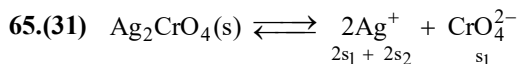
$$[\text{H}^+]^2 [\text{S}^{2-}] = 10^{-22}$$

$$\text{or } [\text{H}^+]^2 = \frac{10^{-22}}{[\text{S}^{2-}]} = \frac{10^{-22}}{10^{-19}} = 10^{-3} = 10 \times 10^{-4}$$

$$\text{or } [\text{H}^+] = (10)^{1/2} \times 10^{-2} \text{ (M)} \quad \text{or } \log [\text{H}^+] = \frac{1}{2} \log 10 - 2$$

$$\text{or } \text{pH} = 2 - \frac{1}{2} \log 10 = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\therefore (\text{minimum pH}) \times 10 = 3/2 \times 10 = 15$$



$$\therefore 4(s_1 + s_2)^2 \times s_1 = 9 \times 10^{-12}$$

$$4(s_1 + s_2)^2 \times s_2 = 6 \times 10^{-12}$$

$$\text{or } \frac{s_1}{s_2} = 3/2 \quad s_2 \quad \text{or } s_1 = 3/2 s_2 \quad \therefore 4 \times \left(\frac{5}{2}\right)^2 \times s_2^3 = 6 \times 10^{-12}$$

$$\text{or } s_2^3 = \frac{6 \times 10^{-12}}{25} = 24 \times 10^{-14} = 240 \times 10^{-15}$$

$$\text{or } s_2 = (240)^{1/3} \times 10^{-5} = 6.214 \times 10^{-5} \text{ (M)}$$

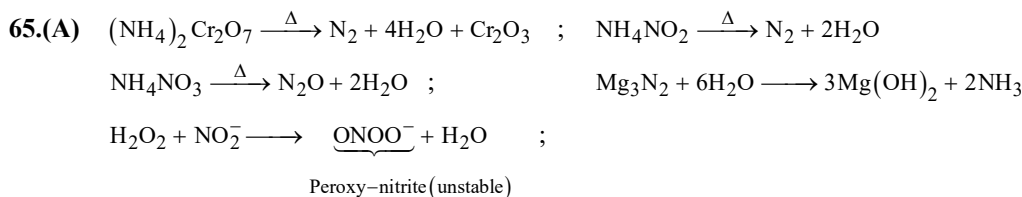
$$s_1 = 6.214 \times 10^{-5} \times \frac{3}{2} = 9.321 \times 10^{-5} \text{ (M)}$$

$$\begin{aligned}\therefore [\text{Ag}^+] &= 2(6.214 + 9.321) \times 10^{-5} \\ &= 31.07 \times 10^{-5} \\ \text{or } [\text{Ag}^+] \times 10^5 &= 31.07 \approx 31\end{aligned}$$

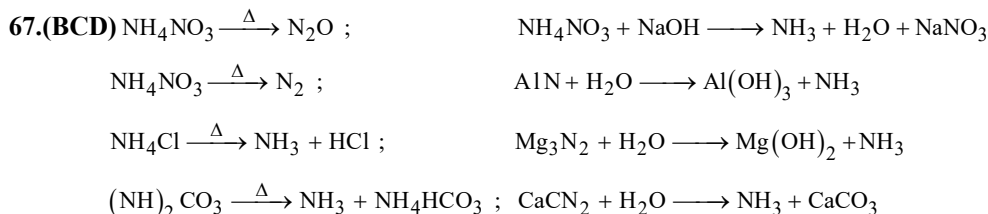
Hydrogen, s & p-Block Elements & Compounds

- 1.(AB) On combustion with excess of air, sodium forms peroxide, Na_2O_2 and some superoxide, NaO_2 . (Read NCERT)
- 2.(D) Beryllium does not exhibit coordination number more than four as in its valence shell there are only four orbitals. (Read NCERT).
- 3.(D) Read NCERT (S-block Elements)
- 4.(C) Which of the following shows an increase in solubility down the group : [Read option (A) as follows]
(A) Alkali metals hydroxide
- 5.(B) More the electropositive character of the cation, more will be the thermal stability. (Read NCERT, S-Block Elements)
- 6.(A) Clearly O_2^- (in KO_2) has odd number electrons and hence will be paramagnetic. Check yourself about the diamagnetic nature of O_2^{2-} (in Na_2O_2 & PbO_2) from molecular orbital theory and of O_3 through its Lewis structure.
- 7.(ABC) Remember the correct statements as fact.
- 8.(C) $\text{Na}_2\text{O} + \text{H}_2\text{O} \longrightarrow 2\text{NaOH}$ $\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$
 $\text{2KO}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{NaOH} + \text{H}_2\text{O}_2 + \text{O}_2$
 $\downarrow$
 $2\text{NaOH} + \text{H}_2\text{O} + \frac{3}{2}\text{O}_2$
 $\downarrow$
 $2\text{NaOH} + \text{H}_2\text{O} + \frac{1}{2}\text{O}_2$
- 9.(A) Fact
- 10.(BC) Alkali carbonate (except Li_2CO_3) are V. stable & NH_4^+ behaves as Alkali metal ions.
- 11.(BD) Read NCERT (S-Block Elements)
- 12.(A) Remember as a fact
- 13.(D) Read NCERT (Hydrogen Chapter), Section 9.4.3, page 279.
- 14.(B) Read option (B) as $\text{Ca}(\text{OD})_2$
- 15.(C) Fact : $\rightarrow$ Apart from Borax, microcosmic salt $[\text{Na}(\text{NH}_4)\text{HPO}_4]$ is also for bead test.
- 16.(B) Crude sodium chloride, generally obtained by crystallisation of brine solution, contains sodium sulphate, calcium sulphate, calcium chloride and magnesium chloride as impurities. Calcium chloride, CaCl_2 , and magnesium chloride, MgCl_2 make common salt hygroscopic because they are deliquescent (absorb moisture easily from the atmosphere). [Read NCERT (s-Block Elements)]
- 17.(D) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2 \uparrow$ (Colourless gas) ; $\text{CaO} + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{OH})_2$;
 $\text{Ca}(\text{OH})_2 + \text{CO}_2(\text{excess}) \longrightarrow \text{Ca}(\text{HCO}_3)_2$; $\text{Ca}(\text{HCO}_3)_2 \xrightarrow{\Delta} \text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2$
- 18.(D) Read NCERT (s-Block Elements)
- 19.(C) $2\text{Ca}(\text{OH})_2 + 2\text{Cl}_2 \longrightarrow \text{CaCl}_2 + \text{Ca}(\text{OCl})_2 + 2\text{H}_2\text{O}$
- 20.(D) Fact
- 21.(AC)

- 22.(C) ➤ Magnesium is a highly flammable metal, but, while it is easy to ignite when powdered or shaved into thin strips, it is difficult to ignite in mass or bulk. Once ignited, it is difficult to extinguish, being able to burn in nitrogen (forming magnesium nitride), carbon dioxide (forming magnesium oxide, and carbon), water (forming magnesium oxide and hydrogen) and sulphur dioxide (forming magnesium oxide, and sulphur).
- Also Read NCERT (S-Block Elements), section 10.6.6.
- 23.(D) Read NCERT (S-Block Elements), Section 10.1.6 24.(BD) Read NCERT (Hydrogen Chapter), Section 9.6.6
- 25.(D) Alums crystallize in regular octahedral (fact). 26.(B) 27.(D)
- 28.(D) NCERT (Hydrogen), Section 9.5.2/Page 280 29.(C) 30.(B)
- 31.(A) NCERT (Hydrogen), Section 9.5.3/Page 281 32.(A) NCERT (Hydrogen), Section 9.6.7/Page 284
- 33.(A) NCERT (Hydrogen), Section 9.5.3/Page 281 34.(CD) NCERT (Hydrogen), Section 9.1/Page 277
- 35.(AB) NCERT (Hydrogen), Section 9.3.2/Page 278
- 36.(AC) NCERT (Hydrogen), Section 9.8/Page 286 & table 9.3/Page 282
Fact : Hydrogen bonds with deuterium are slightly stronger than the ones involving normal hydrogen.
- 37.(AB)
- 38.(CD) NCERT (Hydrogen)/Section 9.6.1, 9.6.2 & 9.6.3 (Page 281-283)
Fact : Water though dissolves ionic compounds and partially covalent compounds but still it is called as “universal solvent” as it dissolved more substances than any other liquid.
- 39.(AB) “Hardness” from NCERT (Hydrogen)
- 40.(BC) NCERT (Hydrogen), Section 9.5.2/Page 280 41.(AD) NCERT(Hydrogen), Section 9.5.2, Page 280
- 42.(C) 43.(A) NCERT (s-Block), Section 10.1.6/Page 294 44.(C) NCERT (s-Block), Section 10.1.6, Page 294
- 45.(AB) NCERT (s-Block), Section 10.7 (Page 301), & Section 10.3.2 Page 278
- 46.(B) NCERT (s-Block), Section 10.2.2, Page 295 47.(A) NCERT (s-Block), Section 10.6.6, Page 300
- 48.(D) NCERT (s-Block), Section 10.7, Page 301 49.(A) NCERT (s-Block), Section 10.2.3, Page 295
- 50.(A) NCERT (s-Block), Section 10.7, Page 301
- 51.(BD) NCERT (s-Block), Section 10.1.2 & 10.1.4 Page 292
- 52.(AC) NCERT (s-Block), Section 10.4 53.(A) NCERT (s-Block), Section 10.7, Page 301
- 54.(D) Volume strength of $\text{H}_2\text{O}_2 = N_{\text{H}_2\text{O}_2} \times 5.6 = 1 \times 5.6 = 5.6 \text{ V}$
- 55.(AC) NCERT (s-Block), Section 10.7, Page 301 56.(AB) NCERT (s-Block), Section 10.8, Page 302
- 57.(ABC) NCERT (s-Block Elements), Section 10.3, Page 296
- 58.(B) Fact : Silica & diamond are electrical insulators, however graphite conducts electricity 59.(B)
- 60.(D) ➤ Recall ‘Inert pair effect’. ➤ Read NCERT (p-Block Elements-XIth)
Section 11.6, Page 315
- 61.(C) Read NCERT (p-Block Elements-XIth), Section 11.8.4, Page 321
- 62.(D) Read NCERT (p-Block Elements), Section 11.3.1, Page 312 63.(B)
- 64.(ABD) $\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{HCl}} \text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{B}_2\text{O}_3 \xrightarrow[\Delta(\text{Reduction})]{\text{Mg/Al/C}} \text{B}$



66.(D)



68.(AD) **69.(A)** Read XIIth NCERT (p-Block Elements), Section 7.5 (Page 174) **70.(C)**

71.(D) Lower the oxidation state, the better is the reducing character. **72.(C)** Fact **73.(D)**

74.(D) Read XIIth NCERT (p-Block Elements), Section 7.5 (Page 175) **75.(B)**

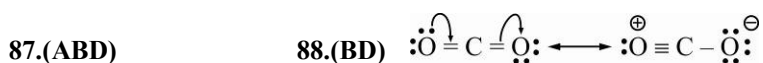
76.(B) Read XIIth NCERT (p-Block Elements), Section 7.5 (Page 175) **77.(A)** **78.(A)**

79.(B) XIth NCERT (p-Block), Section 11.6/Page 315 **80.(B)** Read NCERT.

81.(A) XIth NCERT (p-Block), Section 11.8.6/Page 322 **82.(B)** XIth NCERT (p-Block), Section 11.8.4/Page 322

83.(AB) XIth NCERT (p-Block), Section 11.1.2/Page 309 **84.(B)** **85.(AB)**

86.(BC) ➤ NCERT XIth (p-Block), Section 11.7.2 and 11.7.3 (Page 318)
 ➤ Fullerenes don't have dangling bonds ➤ Graphite is slippery and soft.



89.(C) **90.(A)** **91.(A)** For isoelectronic species, we count valence electrons.

92.(B) NCERT (Solids), Section, 1.3.4 (Page 5) and Table 1.2 (Page 6) **93.(C)**

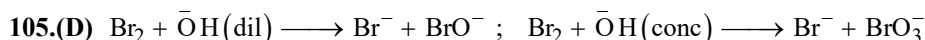
94.(C) NCERT XIth (Hydrogen), Section 9.3.2 **95.(A)** NCERT XIIth (p-Block), Section 7.1

96.(D) NCERT XIIth (p-Block), Section 7.14 (Page 187)

97.(C) O^{2-} donates electron pair to H^+ (from $\text{H}-\text{OH}$) **98.(D)** **99.(B)**

100.(C) Read NCERT XIIth (p-Block Elements), Section 7.1.7 (Page 168) **101.(A)** **102.(C)**

103.(B) **104.(C)**



106.(C) Recall inert pair effect. **107.(D)** **108.(C)** Recall Fajan's rules.

109.(C) **110.(D)** Read NCERT (p-Block Elements)

111.(C) Read NCERT (p-Block Elements) **112.(A)**

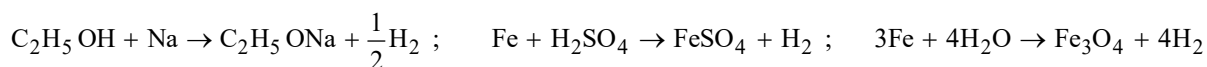
113.(C) Read XIIth NCERT (p-Block Elements), Section 7.23.5 Page 205 uses

114.(B)

115.(C)

116.(A) Heavy water is used for slowing down the speed of neutrons in nuclear reactors, hence used as moderators. Boiling point of heavy water is greater (374.42 K) than that of ordinary water (373 K), hence heavy water is more associated. Dielectric constant of ordinary water is greater than that of heavy water, hence ordinary water is a better solvent.

117.(A) Copper is a noble metal, as it lies below hydrogen in the electrochemical series. Therefore it can't displace hydrogen from dilute HCl. While iron and sodium lie above H in the electrochemical series, so they can liberate H_2 either from steam or H_2SO_4 solution.



118.(B) Hardness of water, due to the presence of chlorides and sulphates of Ca and Mg is called permanent hardness. Hence hard water will consist of Mg^{2+} and Cl^- ions.

119.(B) When ice melts, its molecules move into the holes or open spaces and comes closer to each other than they were in solid state. Thus, ice has lower density than water and there is contraction in volume.

120.(B) $\begin{array}{c} H \\ | \\ O-O \\ | \\ H \end{array}$ is the true structure.

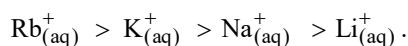
121.(C) Solubility of alkaline earth metal sulphates decreases down the group because hydration energy decreases.

122.(B) The hydration enthalpy of $BeSO_4$ is higher than its lattice energy. Within group 2, the hydration energy decreases down the group while lattice energy is almost the same.

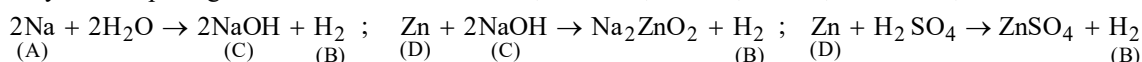
123.(C) Alkali metals are highly electropositive and halogens are electronegative. Thus for the halides of a given alkali metal, the covalent character decreases with increase in electronegativity of halogens.

$\therefore$ Order of covalent character of halides is : $MI > MBr > MCl > MF$.

124.(A) The alkali metal ions exist as hydrated ions $M^+(H_2O)_x$ in the aqueous solution. The degree of hydration, however, decreases with the ionic size as we move from Li^+ to Cs^+ . In other words, Li^+ ion is most highly hydrated. e.g. $[Li(H_2O)_6]^+$. Since the mobility of ion is inversely proportional to the size of their hydrated ions, therefore, amongst the alkali metal ions, lithium has the lowest ionic mobility.

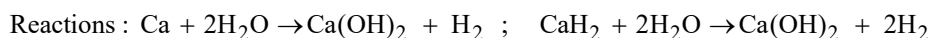


125.(C) Only 'Na' imparts golden colour to Bunsen flame, therefore, A = Na, B = H_2 , C = NaOH, D = Zn.

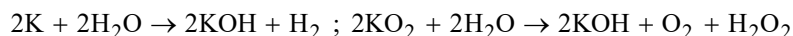


126.(B) As we move down the group from $BeSO_4$ to $BaSO_4$ the enthalpy of hydration of the positive ion becomes smaller due to increase in ionic size. Salts of heavier metal ions are less soluble than those of lighter ions.

127.(C) The pair which gives the same gaseous product is Ca and CaH_2 .



K gives H_2 while KO_2 gives O_2 and H_2O_2 .



Na gives H_2 , while Na_2O_2 gives H_2O_2 .



Ba gives H_2 while BaO_2 gives H_2O_2 .



128.(D) The alkali metals are larger in size and have smaller nuclear charge thus they have low ionization energy in comparison to alkaline earth metals. The alkaline earth metals have fully filled (ns^2) s-orbital and have more nuclear charge.

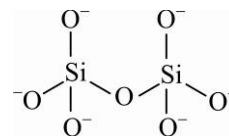
29.(A) In group 13 elements, stability of +3 oxidation state decreases down the group while that of +1 oxidation state increases due to inert pair effect.

Hence, stability of +1 oxidation state increases in the sequence : $Al < Ga < In < Tl$.

130.(A) Me_3SiCl can form only dimer, whereas other can form polymers.

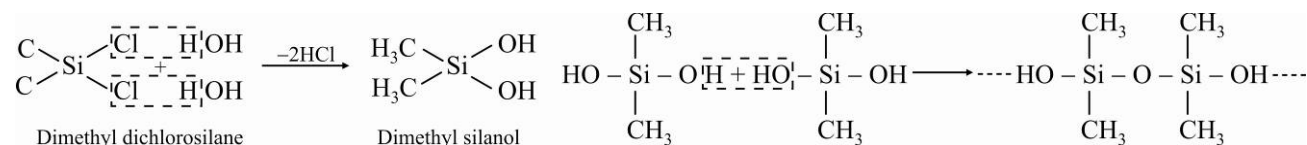
131.(D) SiO_4^{4-} ortho-silicate is basic unit of silicates.

132.(C) Pyrosilicate contains two units of SiO_4^{4-} joined along a corner containing oxygen atom.



133.(ABD) Al reacts with NaOH to give sodium meta-aluminate.

134.(C) Hydrolysis of substituted chlorosilanes yields corresponding silanols which undergo polymerisation. Out of the given chlorosilanes, only $(CH_3)_2SiCl_2$ will give linear polymer on hydrolysis followed by polymerisation.



135.(D) $SeO_2 \rightarrow$ acidic oxide ; $Al_2O_3 \rightarrow$ amphoteric ; $Sb_2O_3 \rightarrow$ amphoteric ; $Bi_2O_3 \rightarrow$ basic oxide

136.(B) There is no lone pair on boron in BCl_3 hence no repulsion place. There is a lone pair on nitrogen in NCl_3 hence repulsion takes place. Therefore BCl_3 is planar molecule but NCl_3 is pyramidal molecule.

137.(A) Carbon has no d-orbitals, while silicon contains d-orbitals in its valence shell which can be used for bonding purpose.

138.(C) H_3BO_3 is a weak monobasic acid. We know that $B(OH)_3 + H_2O \rightarrow [B(OH)_4]^- + H^+$. Therefore it is a weak monobasic acid.

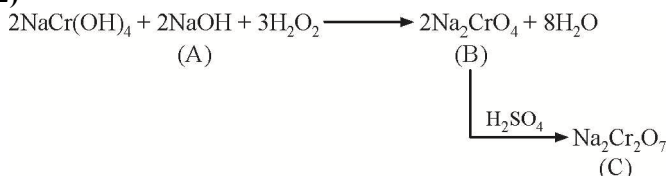
139.(40) 48 mg of $MgSO_4$ present in 10^3 g of water

$$10^6 \text{ g water contains } MgSO_4 = 48 \times 10^3 \\ = 48 \text{ g}$$

$$\therefore 120 \text{ g of } MgSO_4 \equiv 100 \text{ g of } CaCO_3$$

$$\Rightarrow 48 \text{ g } MgSO_4 \equiv \frac{100}{120} \times 48 = 40 \text{ ppm}$$

140.(262)



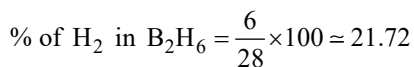
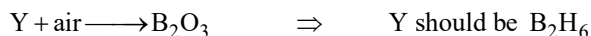
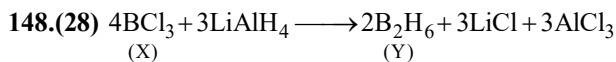
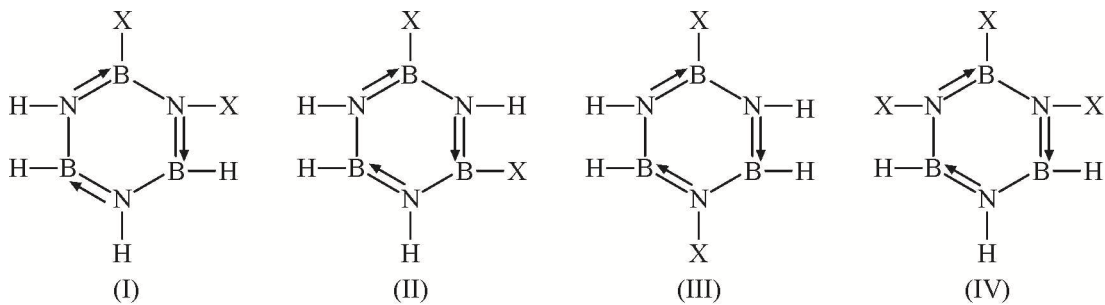
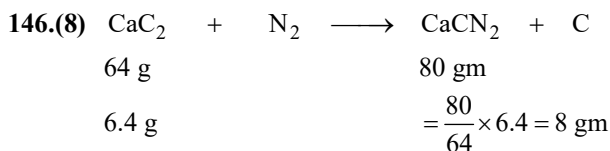
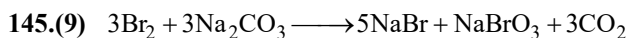
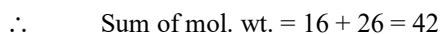
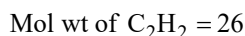
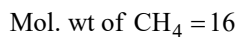
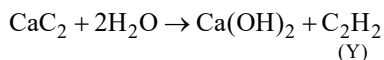
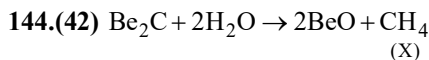
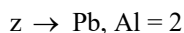
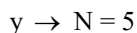
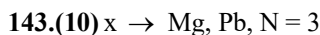
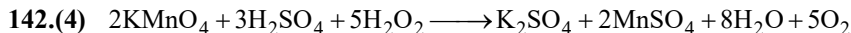
$$\text{Mol wt of } Na_2Cr_2O_7 = (23 \times 2) + (52 \times 2) + (16 \times 7)$$

$$= 46 + 104 + 112$$
$$= 262$$

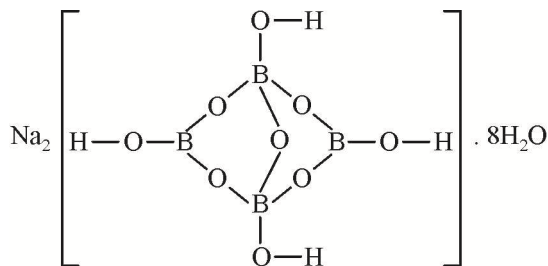
141.(5) $N_1 = \frac{10}{5.6}$, $N_2 = \frac{15}{5.6}$, $N_3 = \frac{20}{5.6}$

$$\left(\frac{10}{5.6} \times \frac{1}{2}\right) + \left(\frac{15}{5.6} \times \frac{1}{2}\right) + \left(\frac{20}{5.6} \times \frac{1}{2}\right) = N_T \times 4.5$$
$$\Rightarrow N_T = 0.8928 \text{ N}$$

Volume strength = $5.6 \times 0.8928 \approx 5$



149.(68) $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

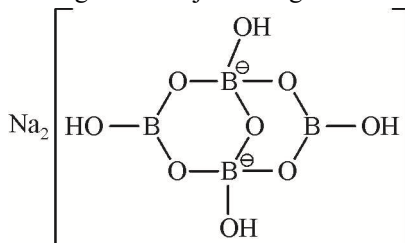


$$\text{L.p} = 17 \times 2 = 34$$

$$\text{B.p} = 34$$

$$\text{Total} = 34 + 34 = 68$$

150.(2) Borax has two tetrahedral and two triangular units joined together.



151.(6) $(\text{Si}_2\text{O}_7)^{6-}$

152.(4) Clark's method is used for removal of temporary hardness.

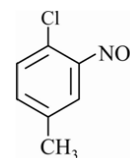
153.(3) Superoxide (O_2^-) ion is stable only in the presence of large cations such as K, Rb, Cs etc.

IOC & GOC

1.(A) Numbering of carbon chains follows lowest locant rule whereas sequence of writing groups chain follows alphabetical order.

2.(D) Priority of $-\text{COOH}$ group is more than $-\text{CO}-$ group, so numbering starts from carbonyl carbon atom in the carboxylic group. Thus name is 4-oxopentanoic acid.

3.(B) In aromatic compounds, numbering starts from the carbon atom at which group with alphabetical priority is attached. Out of $-\text{Cl}$ (chloro), $-\text{NO}_2$ (Nitro) and $-\text{CH}_3$ (methyl) groups, $-\text{Cl}$ has highest priority. After that numbering follows in such a way that other group get lowest number.



1-chloro-3-methyl-2-nitrobenzene

4.(C) The percentage of s-character in hybridization increases electronegativity of carbon atom. In $\text{CH}_3-\text{CH}_2-\text{C}^*\equiv\text{CH}$, said carbon atom is sp -hybridized, so, it most electronegative due to 50% (highest) s-character.

5.(ABD)

Functional isomerism is possible for compounds containing functional group. Alkyl halides cannot show functional isomerism since these are side chain (on groups).

- 6.(A) Resonance dominates hyperconjugation, so carbocation containing resonance effects is more stable.



$\text{CH}_3 - \overset{+}{\text{C}}\text{H} - \text{CH}_3$ contains six α -hydrogen and $\text{CH}_3 \overset{+}{\text{C}}\text{H} - \text{CH}_2 - \text{OCH}_3$ contains five α -hydrogen,

So, $\text{CH}_3 \overset{+}{\text{C}}\text{HCH}_3$ (I) is more stable than (III) so, correct order is $\text{II} > \text{I} > \text{III}$.

- 7.(B) Longest carbon chain is six carbon atoms, 3rd & 4th position is occupied by methyl groups. Thus IUPAC name is 3, 4-Dimethylhexane.

- 8.(A) In Grignard reagent, $\text{CH}_3\text{CH}_2\text{MgCl}$, C – Mg bond is ionic bond due to which carbon atom is negativity charged and it shows +I effect. –I group will make carbon atom more positively charged. Thus $\text{CH}_3\text{CH}_2\text{Cl}$ contains more positively charged carbon atom due to more –I effect of –Cl.

- 9.(D) Carboxylate ion is most stable of it is formed by deprotonation of strongest conjugate acid.

Acidic strength : $\text{F}_2\text{CHCOOH} > \text{FCH}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{CH}_3\text{COOH}$

Stability of ion : $\text{F}_2\text{CHCOO}^- > \text{FCH}_2\text{COO}^- > \text{ClCH}_2\text{COO}^- > \text{CH}_3\text{COO}^-$

- 10.(C) $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{H}^+} \text{CH}_3 - \overset{+}{\text{C}}\text{H} - \text{CH}_3$ (2° Carbocation)

- 11.(AD) $\text{H} - \text{C} \equiv \text{C} - \text{C} \equiv \text{C} - \text{H}$, only sp-hybridized carbon atoms.

$\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$, only sp^2 -hybridized carbon atoms.

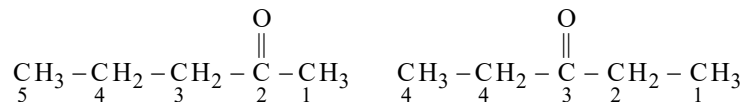
- 12.(BC) There should be an empty orbital to accept electron pair, for a species to be electrophile.

AlCl_3 – p-orbital, SO_3 – d-orbital empty

NO_2^+ – p-orbital empty

$\overset{+}{\text{C}}\text{H}_3$ – p-orbital empty, $\text{CH}_3 - \overset{+}{\text{C}} = \text{O}$ – p-orbital empty

- 13.(B) For positive isomerism, parent carbon chain should contain same carbon atoms.



Only position of ketonic group is different in the compounds

- 14.(AC) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$ $\text{CH}_3\text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2\text{CH}_3$

Position isomers.

$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$

$(\text{CH}_3)_2\text{CH} - \text{CH}_2 - \text{CHO}$

Different Chain, some functional group

$\Rightarrow$ Chain isomers.

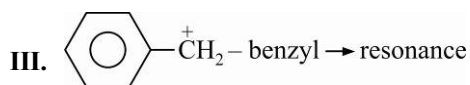
- 15.(A) Answer can be either (A) or (C) but the question says “Nucleophile is species that **should have**”. So, it should have a pair of electrons to donate. It may or may not have a negative charge.

- 16.(AB) Hyperconjugation observed in two cases.

* Delocalization of σ -electrons into empty p-orbital * Delocalization of σ -electrons into empty π^* orbitals.

- 17.(B) Both statements are true but reason is not correct explanation of A.

18.(A) I. $\text{CH}_2 = \text{CH} - \overset{+}{\text{C}}\text{H}_2 - \text{Alkyl} \rightarrow \text{resonance}$ II. $\text{CH}_3 - \overset{+}{\text{C}}\text{H} - \text{CH}_3 - \text{isopropyl} \rightarrow \text{only hyperconjugation}$

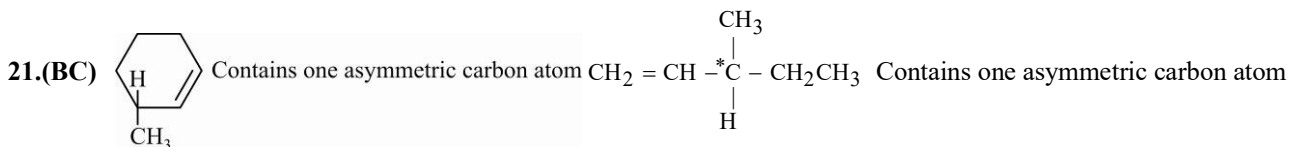
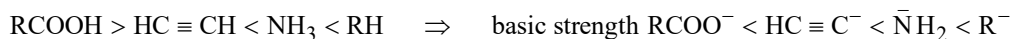


Correct order should be $\text{III} \approx \text{I} > \text{II}$

However, In case of comparison between I and III, usually we take $\text{III} > \text{I}$.

19.(B) RSH is more than ROH but RS^- is more nucleophilic because sulphur atom contains more diffused electron density, so, approachability of RS^- at back side of $\text{C}-\text{X}$ bond is more due to which it is stronger nucleophile.

20.(D) Basicity of given anions is reverse of acidity of their conjugate acids. Acidic order of conjugate acids

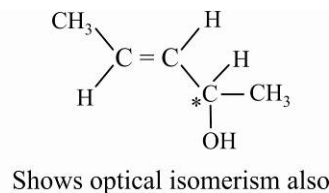
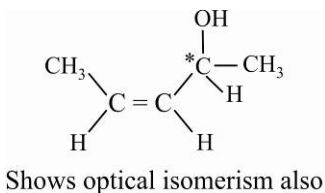


22.(C) Electron withdrawing group stabilize the carbanions $\bar{\text{C}}\text{Cl}_3 \rightarrow \text{very high } -\text{I effect} : \text{C}_6\text{H}_5\bar{\text{C}}\text{H}_2 \rightarrow \text{Resonance effect}$

Thus order is $\bar{\text{C}}\text{Cl}_3 > \text{C}_6\text{H}_5\bar{\text{C}}\text{H}_2 > (\text{CH}_3)_2\bar{\text{C}}\text{H} > (\text{CH}_3)_3\text{C}^-$

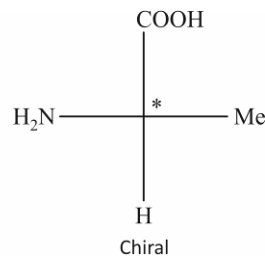
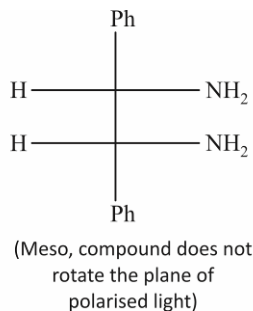
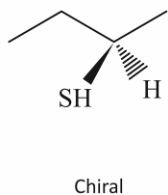
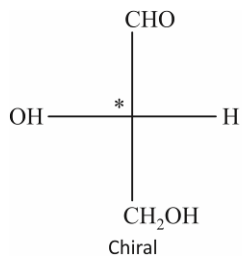
23.(C) Geometrical isomers

Thus total of 4-stereoisomers are possible.

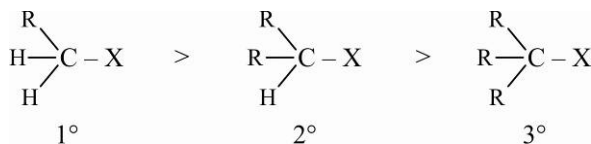


24.(A)

25.(ABD) Compound containing chiral carbon atom rotates plane of polarised light. In the case when more than one chiral centres exist, overall chirality should exist.



26.(B) 1° halides are most reactive and 3° halides are least reactive in $\text{S}_\text{N}2$ reaction because steric hindrance increases from 1° to 3° at back side of $\text{C}-\text{X}$ bond.

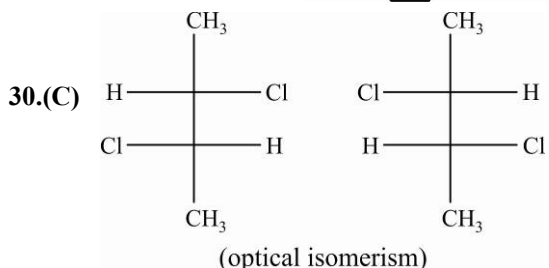
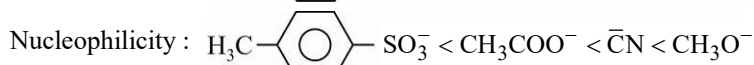
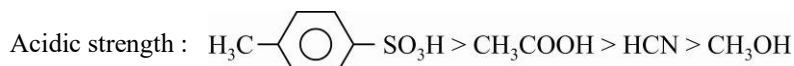


27.(A) Resonance effect containing free radicals are more stable than hyperconjugation effect.



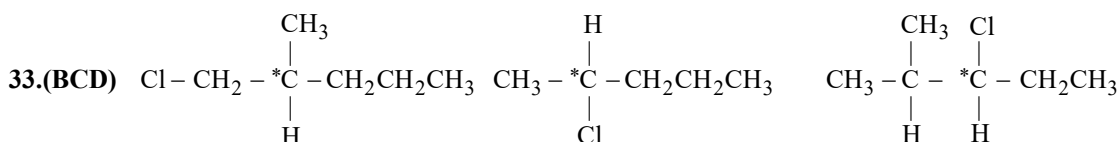
28.(D) Free radicals are chemically reactive species due to unpaired electron.

29.(B) Nucleophilicity is the reverse of the stability of anions, i.e. reverse of the acidity of conjugate acids



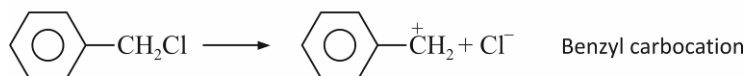
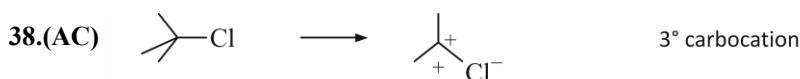
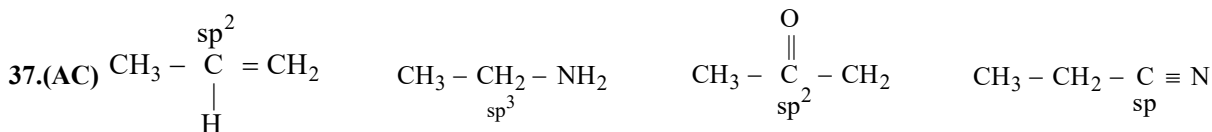
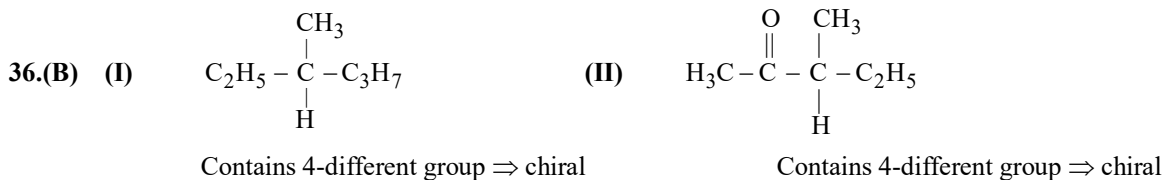
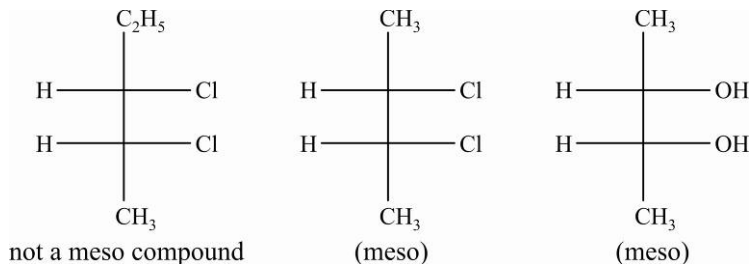
31.(ABD) Acetonitrile is $\text{CH}_3-\text{C}\equiv\text{N}$ which contains sp^3 & sp hybridised carbon atoms. Other compounds contain group in which carbon is sp^2 hybridised carbon atom.

32.(D) In gaseous state, $d \propto M$, $\text{C}_2\text{H}_5\text{OH}$ & CH_3OCH_3 contains same molecular mass, so same density for both.



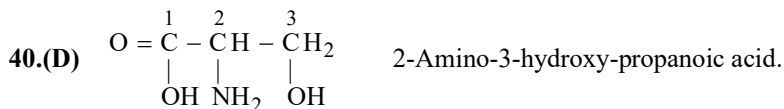
34.(C)

35.(BD) At least two chiral centers should be in the compound with environment around chiral carbon atoms.



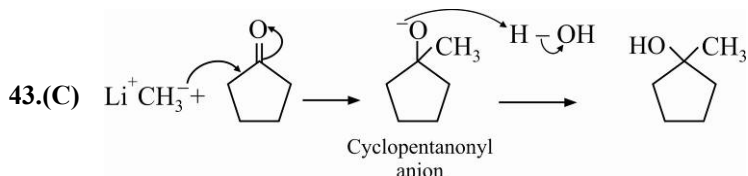
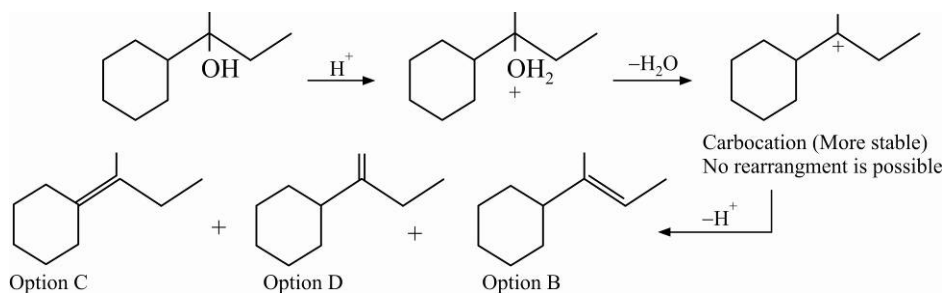
1° carbocation is not formed because stability is very low.

39.(D) Structure I should be treated as 2-butene, hence it will show only geometric isomerism. Structure II should be treated as 2-butanol, hence it will show only optical isomerism. However, it is a single choice question, in that case (D) is correct answer. Both types of isomerism studied under stereoisomerism.



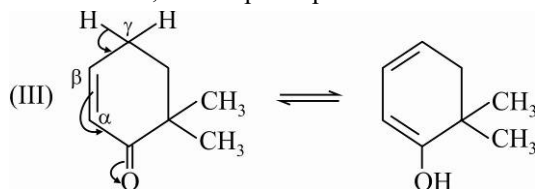
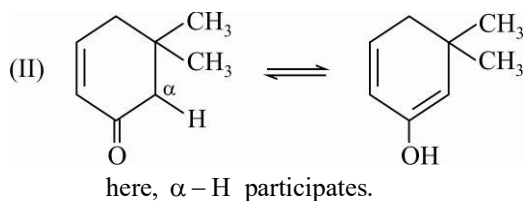
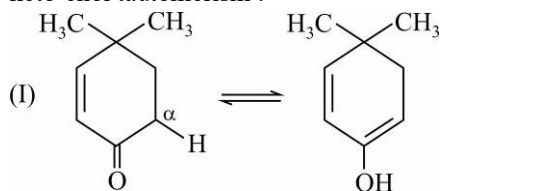
41.(D) Nucleophiles are electron rich species hence, they are Lewis bases.

42.(A)



44.(D) There are four double bonds. Hence, no. of π -electrons = $2 \times 4 = 8$.

45.(B) keto-enol tautomerism :



46.(A) Mass of organic compound = 0.25 g

Experimental values (at STP) :

$$V_1 = 40 \text{ mL} ; V_2 = ? ; T_1 = 300 \text{ K} ; T_2 = 273 \text{ K} ; P_1 = 725 - 25 = 700 \text{ mm} ; P_2 = 760 \text{ mm}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{700 \times 40 \times 273}{300 \times 760} = 33.52 \text{ mL}$$

22400 mL of N₂ at STP weighs = 28 g

$$\therefore 33.52 \text{ mL of N}_2 \text{ at STP weighs} = \frac{28 \times 33.52}{22400} = \frac{33.52}{8} = 0.0419 \text{ g}$$

$$\% \text{ of N} = \frac{\text{Mass of nitrogen at STP}}{\text{Mass of organic compound taken}} \times 100 = \frac{0.0419}{0.25} \times 100 = 16.76\%$$



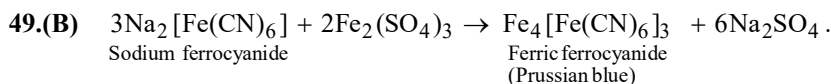
10 mL of 1 M H₂SO₄ = 10 mmol. [$\therefore M \times V_{(\text{mL})} = \text{mmol}$]

NH₃ consumed = 20 mmol

Acid used for the absorption of ammonia = 20 – 10 mmol = 10 mL of 2 N (or 1 M) H₂SO₄

$$\% \text{ of N} = \frac{1.4 \times N \times V}{W} = \frac{1.4 \times 2 \times 10}{0.75} = 37.33\%$$

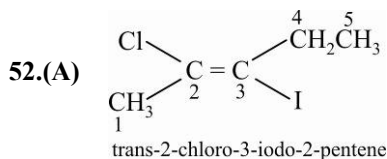
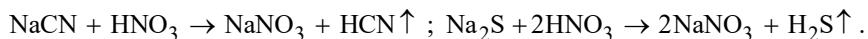
48.(C) Aromatic species has a close loop of delocalizable $(4n + 2)\pi$ electrons. In this species d-electron is not part of close loop. So this species contain 6 delocalizable electrons in 6p-orbitals.



50.(B) Similar to Question 46. Find V_{N_2} .

$$\% \text{ N} = \frac{V_{\text{N}_2} (\text{STP})}{8 \times W_{\text{org. compound}}} \times 100 \Rightarrow \% \text{ of N} = \frac{46.099}{8 \times 0.35} = 16.46$$

51.(A) In case of Lassaigne's test of halogens, it is necessary to remove sodium cyanide and sodium Sulphide from the sodium extract if nitrogen and sulphur are present. This is done by boiling the sodium extract with conc. HNO₃.



53.(A) Ortho-cresol is the most reactive towards electrophilic reagents (in S_E reaction). The reason for this is the –CH₃ group (+I effect) and –OH group (+R/+M effect) which increase the electron density on the benzene ring.

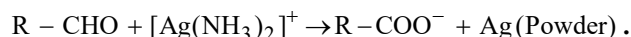
54.(C) H₃O⁺ is not electrophile as it has no vacant orbital.

55.(A) In case of different nucleophiles, but present in the same group in the periodic table, then larger is the size, higher is the nucleophilicity. Hence the decreasing order of nucleophilicity of the halide ions is :
I[–] > Br[–] > Cl[–] > F[–].

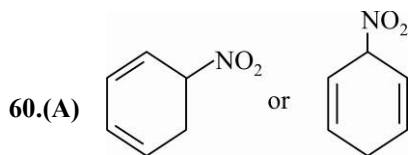
56.(B) Sublimation method is used for those organic substances which pass directly from solid to vapour state on heating and vice-versa on cooling, e.g. benzoic acid, naphthalene, camphor, anthracene, etc. Naphthalene is volatile and benzoic acid is nonvolatile due to the formation of the dimer.

57.(B) Carboxylic acid is much stronger than water and alcohol. Since the carboxylate ion after the removal of proton is stabilised by resonating structures. The –OH in alcohols is almost neutral. Acetylene is also weakest acid.

- 58.(C) Tollen's reagent is solution of ammoniacal silver nitrate and used for the detection of $-\text{CHO}$ group. Aldehydes reduce Tollen's reagent and itself get oxidised to give Ag^+ ions to Ag powder which forms the silver coloured mirror in the test tube. So this test is also known as silver mirror test.

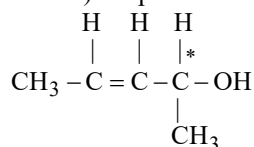


- 59.(D) Carboxylic acid dissolve in NaHCO_3 but phenols do not being very weak acids.

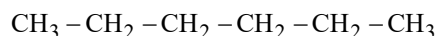


Hence it is homocyclic (as the ring system is made of one type of atoms, i.e., carbon) but not aromatic.

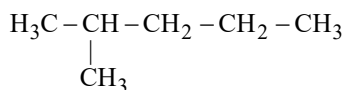
- 61.(4) The compound has one chiral centre. Therefore, $2^n = 2^1 = 2$ optically active compounds are possible. Also, two geometrical isomers (cis and trans) are possible. So, total of four isomers are possible.



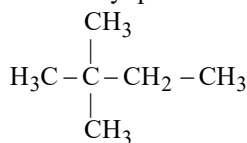
- 62.(5) Structural isomers refers to the compounds that have same molecular formula but different arrangement of atoms or groups within the molecule. Based on this, following are the structural isomers for C_6H_{14} :



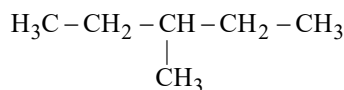
n-Hexane



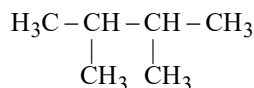
2-Methyl pentane



2,2-Dimethyl butane

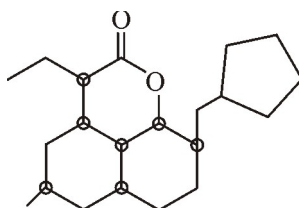


3-Methyl pentane



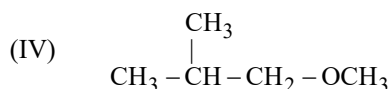
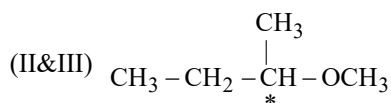
2,3-Dimethyl butane

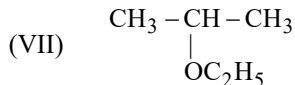
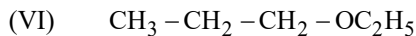
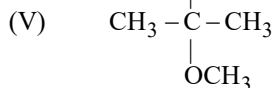
- 63.(7) There are seven chiral centres present in the given compound.



- 64.(8) When molecule is asymmetric, no. of geometrical isomers is given by 2^n where n is no. of double bonds.

- 65.(7) (I) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{OCH}_3$



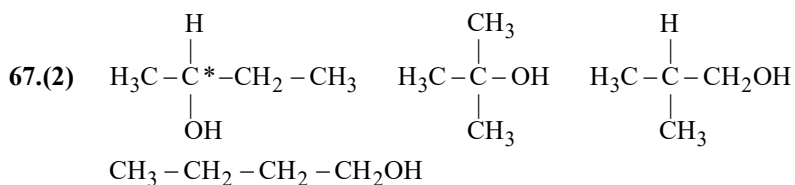


- 66.(11)** Triple bonds = 1

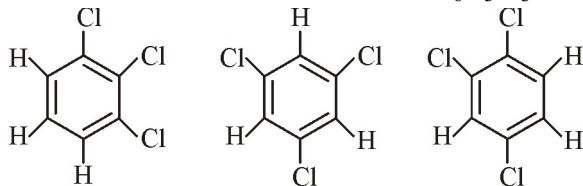
Double bonds = 6

Rings = 3

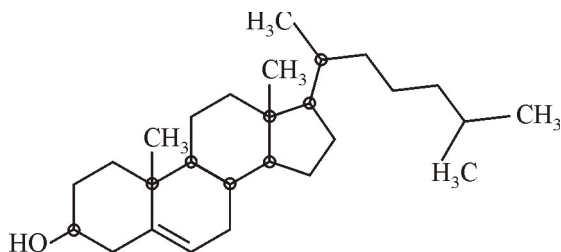
Due to each triple bond, there are 2 degree of unsaturation while due to double bond & ring there is 1 degree of unsaturation each. So in total D.U. = $2 + 6 + 3 = 11$



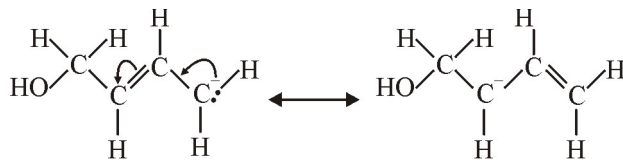
- 68.(3)** There are three derivatives for formula $\text{C}_6\text{H}_3\text{Cl}_3$.



- 69.(8)** There are 8 chiral carbon atoms present.

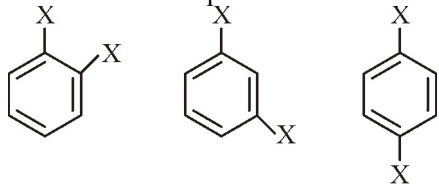


- 70.(2)** The resonating structures are



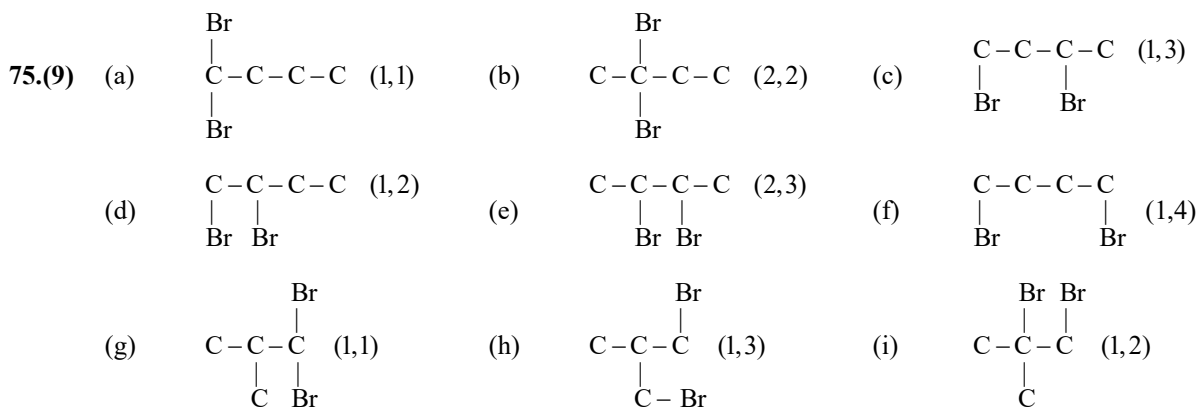
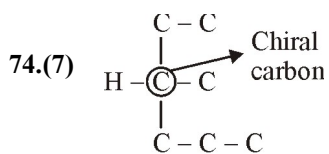
- 71.(6)**
- | | | |
|---|--|-------------------|
| $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH} = \text{CH}_2$ | $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_3$ | |
| 1-Pentene | 2-Pentene | |
| $\begin{array}{c}\text{CH}_3 - \text{C} = \text{CH} - \text{CH}_3 \\ \\ \text{CH}_3\end{array}$ | $\begin{array}{ccc}\text{CH}_3 - \text{CH} - \text{CH} = \text{CH}_2 & & \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ & & \\ \text{CH}_3 & & \text{CH}_2\end{array}$ | |
| 3-Methylbut-2-ene | 3-Methylbutene | 2-Methylbut-1-ene |
- 2-Pentene can show geometrical isomerism also. So in total there are 6 isomers.

72.(3) Three isomers are possible for disubstituted benzene.

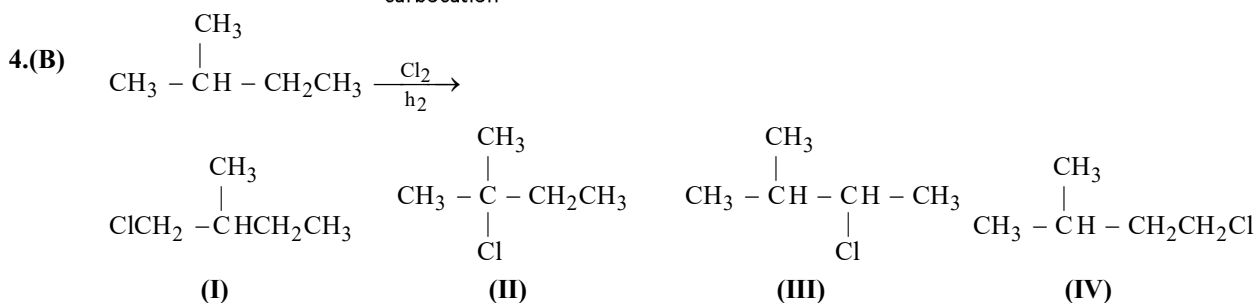
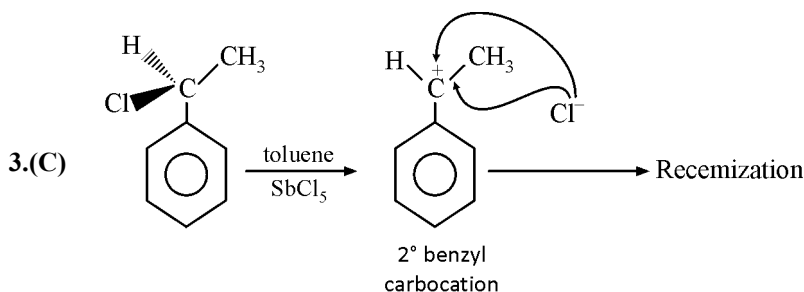
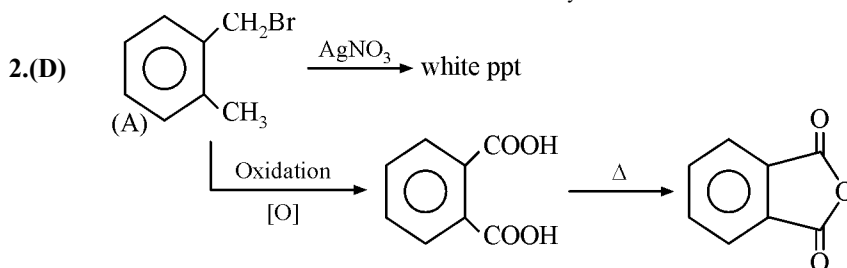
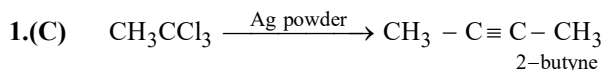


73.(4) $a = 2^n$ where a = number of optically active isomers and n = number of asymmetric carbon atoms

As $n = 2$, so $a = 2^2 = 4$.



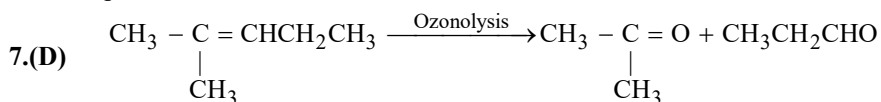
Hydrocarbons



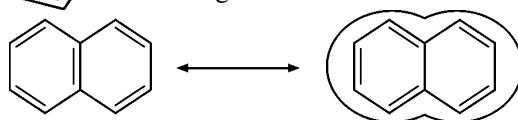
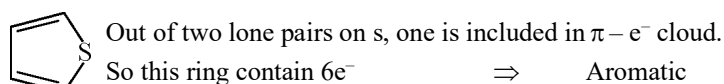
Four structural isomers are produced but (I) and (III) contains chiral centre, so, two isomers are chiral compound,

5.(B) Dissolved metal reduction produces trans isomer and heterogeneous catalyst produces cis isomer

6.(B) $\text{CH}_2 = \text{CH}-$ group should exist on the hydrocarbon which on ozonolysis will produce formaldehyde as one product.

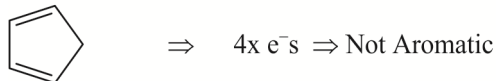


8.(BCD) $(4n + 2) \pi e^-$ rule



Consider one ring and count e^- s on the perimeter of its

It contains $10e^-$ which are according $(4n + 2) \pi e^-$ s $\Rightarrow$ Aromatic

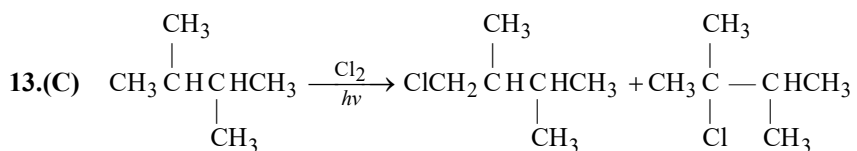
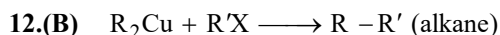


9.(ABCD) All the given alkynes will give corresponding alkene on reaction with sodium in liquid ammonia due to partial hydrogenation of triple bond.

Note : If we use NaNH_2 , then only $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$ and $\text{PhC}\equiv\text{CH}$ containing acidic hydrogen will react.



11.(C)



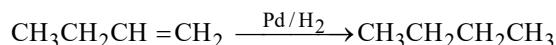
(Only two structural isomers are formed). Stereoisomers are ignored.

14.(C) At lower temperature, thermodynamic controlled products are formed. Thus, 1-bromo-2-butene is formed

15.(D) Zn-Hg/HCl reduces butan-2-one, other compounds requires strong reducing agents

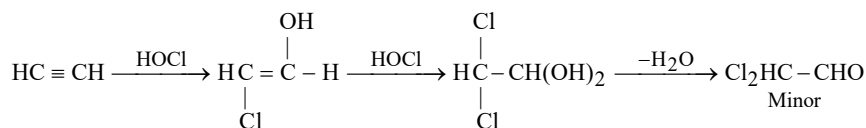
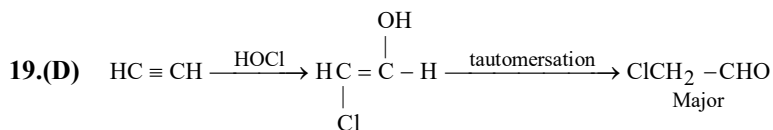
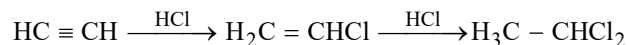
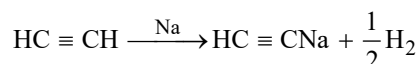
16.(D) Isobutene has lowest surface area, so, contains lowest boiling point.

17.(D) Butene-1 is non polar in nature, so, non-polar catalyst is best to reduce it.



18.(ABC) Acetylene is weakly acidic in nature due to which it cannot react with NaOH .

HCl reacts with triple bond.



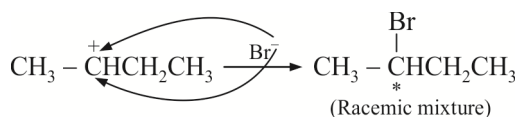
20.(C) CH_3MgI contains $\text{C}-\text{Mg}$ bond

21.(D) n-pentane : Highest molar mass and highest surface area ; n-butane : Lowest molar mass and lowest surface area
n-pentane > 2-methylbutane > 2, 2-dimethylpropane > n-butane

22.(A) $F_2 > Cl_2 > Br_2 > I_2$

23.(B)

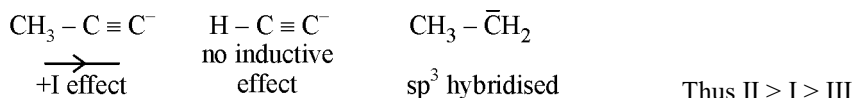
24.(A) $CH_2 = CHCH_2CH_3 \xrightarrow{HBr} CH_3 - \overset{+}{C}HCH_2CH_3 \text{ (major)} + \overset{+}{C}H_2CH_2CH_2CH_3 \text{ (minor)}$



25.(ABC) Each carbon connected by double bond should contain two different group individually.

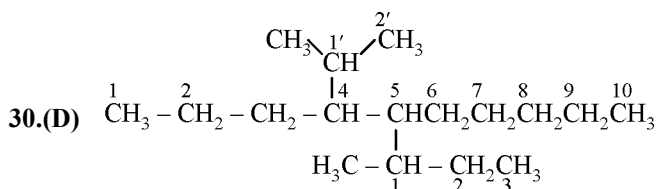
26.(C) Stronger acid reacts faster.

27.(B) sp hybridized carbon atom is more electronegative so, negative charge is more stable on if

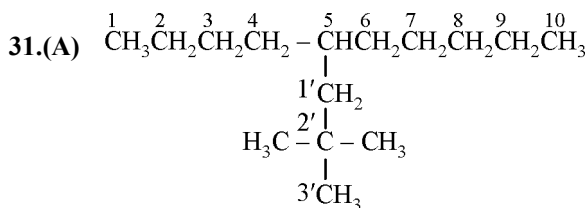


28.(ABC) Factual

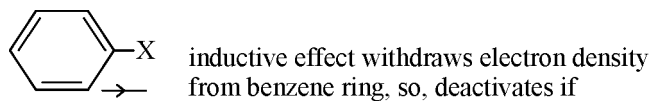
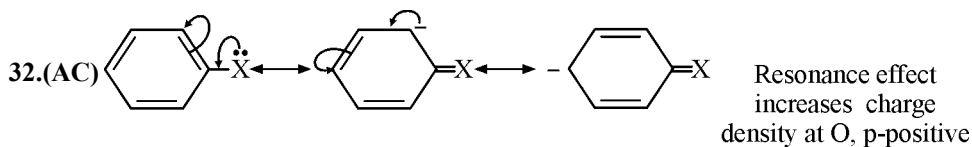
29.(BCD) Factual



4-(1-methylethyl)-5-(1-methylpropyl)-decane



5-(2', 2' - dimethylpropyl) - decane



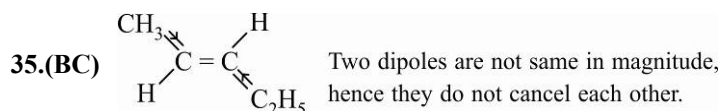
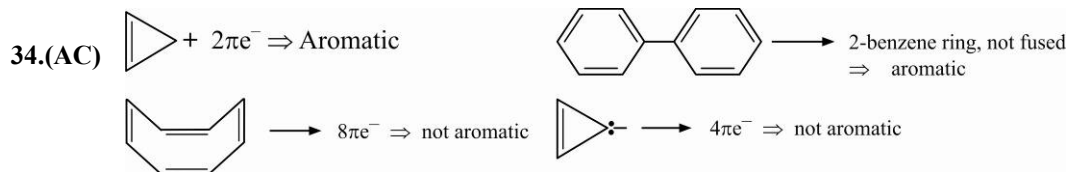
33.(ABC) (A) $CH_3 - \overset{+}{O} - \overset{+}{C}H_3 \longleftrightarrow CH_3 - \overset{+}{O} = CH_2$ Resonance effect

$CH_3 - \overset{+}{C}H_2$ hyperconjugation. Thus $CH_3O\overset{+}{C}H_2$ is more stable than $CH_3\overset{+}{C}H_2$

(B) $CH_3 - \overset{+}{C}H - CH_3$ (2° carbocation) is less stable than $CH_3 - O - \overset{+}{C}H_2$ because of resonance.

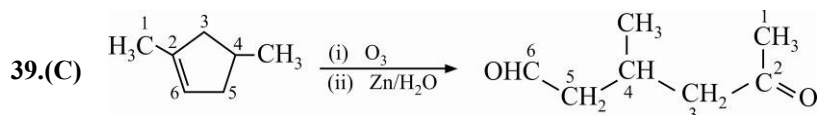
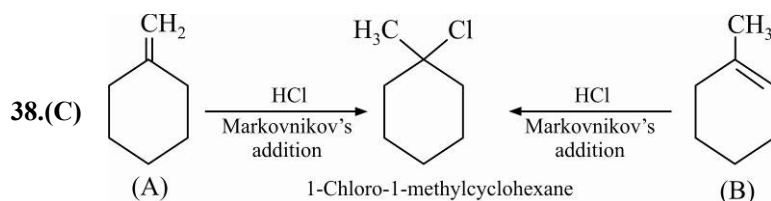
(C) $CH_2 = CH - \overset{+}{C}H_2$ is more stable than $CH_3CH_2 - \overset{+}{C}H_2$ (1° carbocation) because of resonance.

(D) Vinyl carbocation are less stable than normal carbocation $\text{CH}_2 = \overset{+}{\underset{\text{sp}^2}{\text{C}}}\text{H} < \text{CH}_3 - \overset{+}{\text{C}}\text{H}_2$

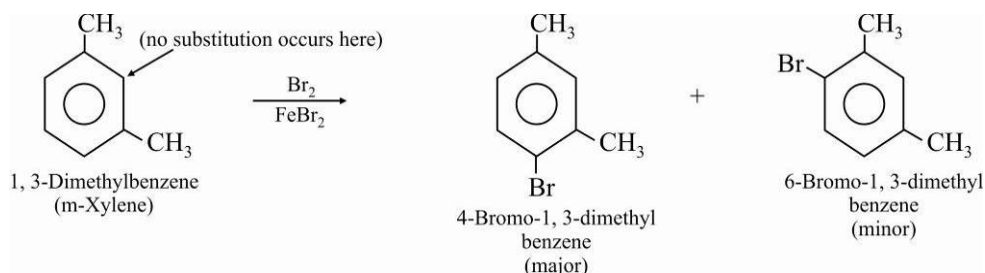


36.(C) Refer to explanation of Q.34

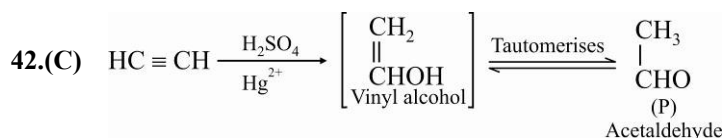
37.(C) More branching means lesser surface area which results in lesser molecular forces of attraction (Van der waals forces) and hence lower boiling point.



40.(C) $-\text{CH}_3$ group is o, p-directing. Because of crowding, no substitution occurs at the carbon atom between the two $-\text{CH}_3$ groups in m-Xylene, even though two $-\text{CH}_3$ groups activate that position.

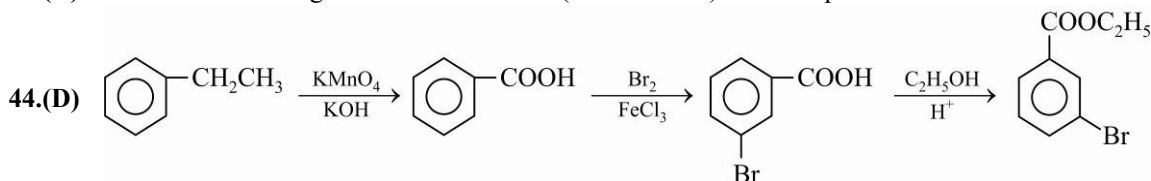


41.(D) The molecules which do not satisfy Huckel rule or $(4n + 2)\pi$ -electron rule are said to be non-aromatic. The compound (d) has total $4\pi e^-$. It does not follow $(4n + 2)$ rule. So it is non-aromatic compound. All other compounds (a, b, c) are planar and have $6\pi e^-$, so they are aromatic.

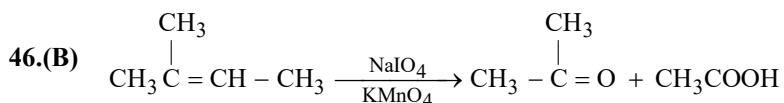


However, acetaldehyde does not give Victor Meyer test. It will give Tollen's test, Brady's test along with iodoform test.

43.(A) Maleic acid shows geometrical isomerism (Fumaric acid) and not optical isomerism.



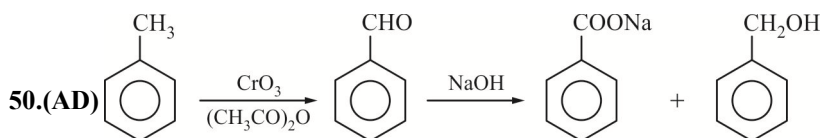
45.(D) The conformation (D) is most stable because of intermolecular H-bonding.



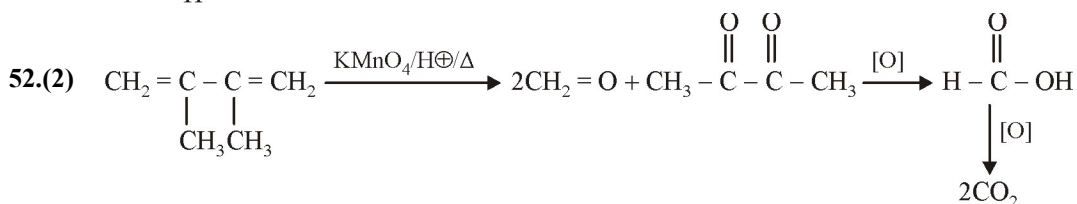
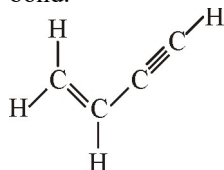
47.(B) Effect of substituents : Presence of electron withdrawing groups ($-\text{NO}_2$, $-\text{X}$, $-\text{CN}$) increase the acidity of phenols while the presence of electron releasing groups ($-\text{NH}_2$, $-\text{CH}_3$) decrease the acidity of phenols. Hence the order of decreasing acidity is : p-nitrophenol > phenol > p-cresol.

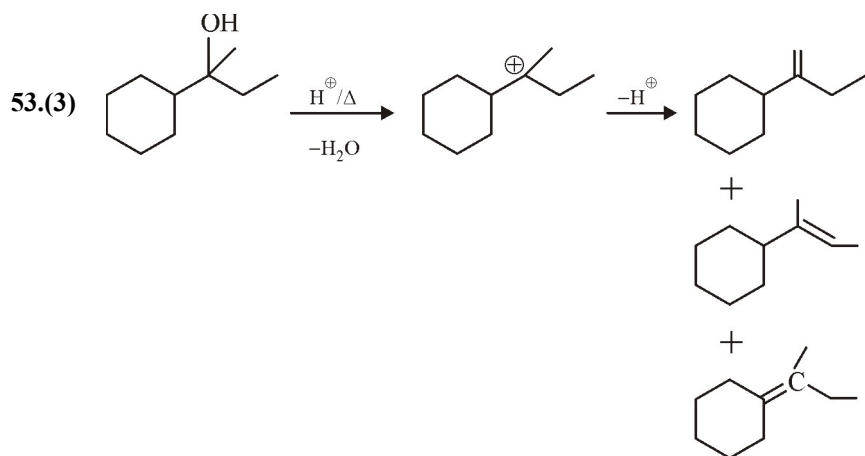
48.(A) Due to $-\text{I}$ effect of F atom, CF_3 in benzene ring deactivates the ring and does not favour electrophilic substitution. While $-\text{CH}_3$ and $-\text{OCH}_3$ via hyperconjugation and resonance effect respectively favour electrophilic substitution in the benzene ring at 'ortho' and 'para' position. The $+\text{M}$ effect of $-\text{OCH}_3$ is more than $+\text{H}$ effect of $-\text{CH}_3$, therefore correct order for electrophilic substitution is (A).

49.(C) The branching of chain increases the octane number of a fuel. High octane number means better fuel.

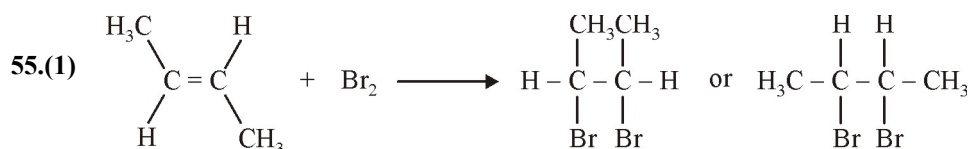
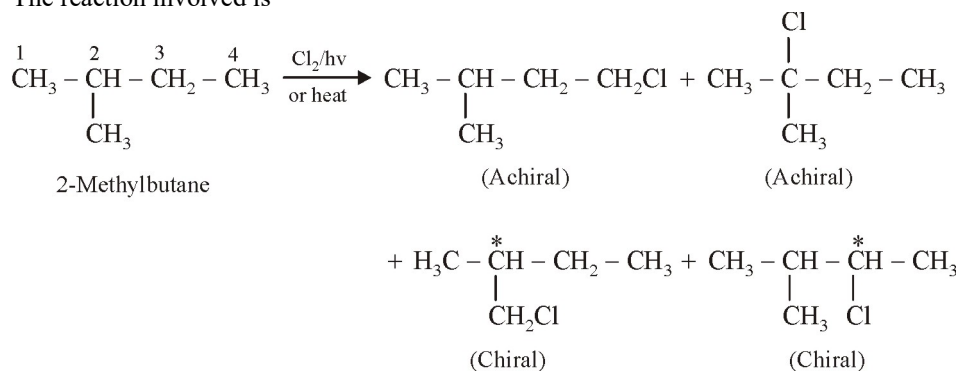


51.(6) One unhybridized orbital on each carbon involved in double bond and two on each carbon involved in triple bond.





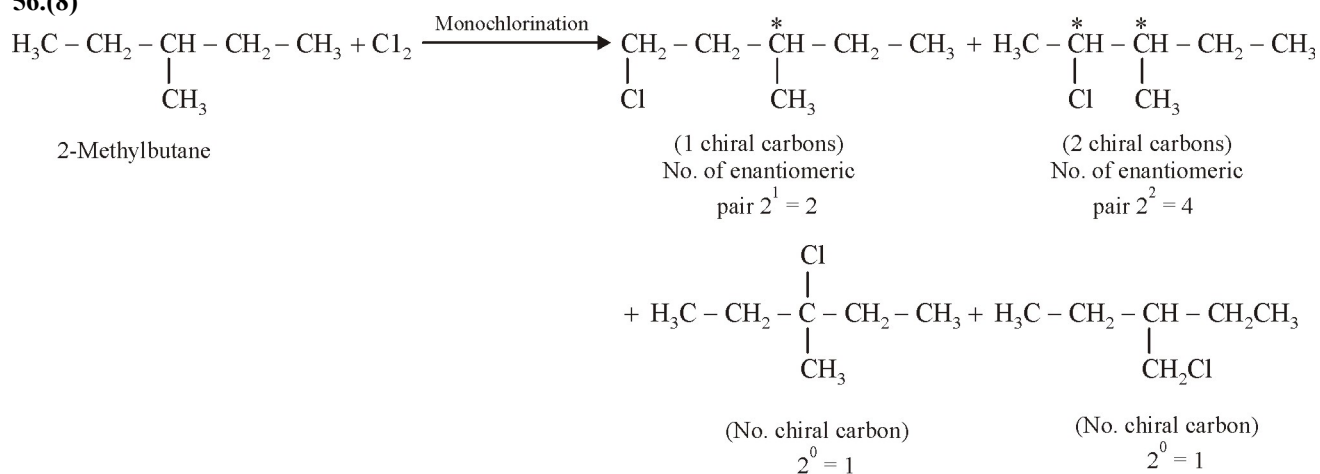
54.(4) The reaction involved is



The compound obtained is a meso compound. Meso compounds are those which are superimposable on their mirror image even though they contain chiral centres.

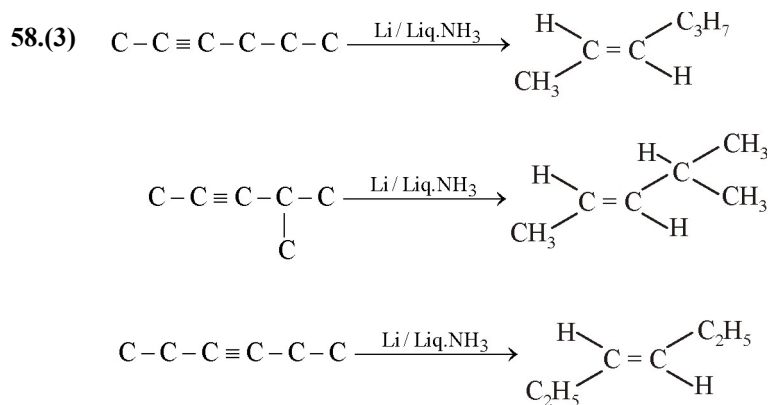
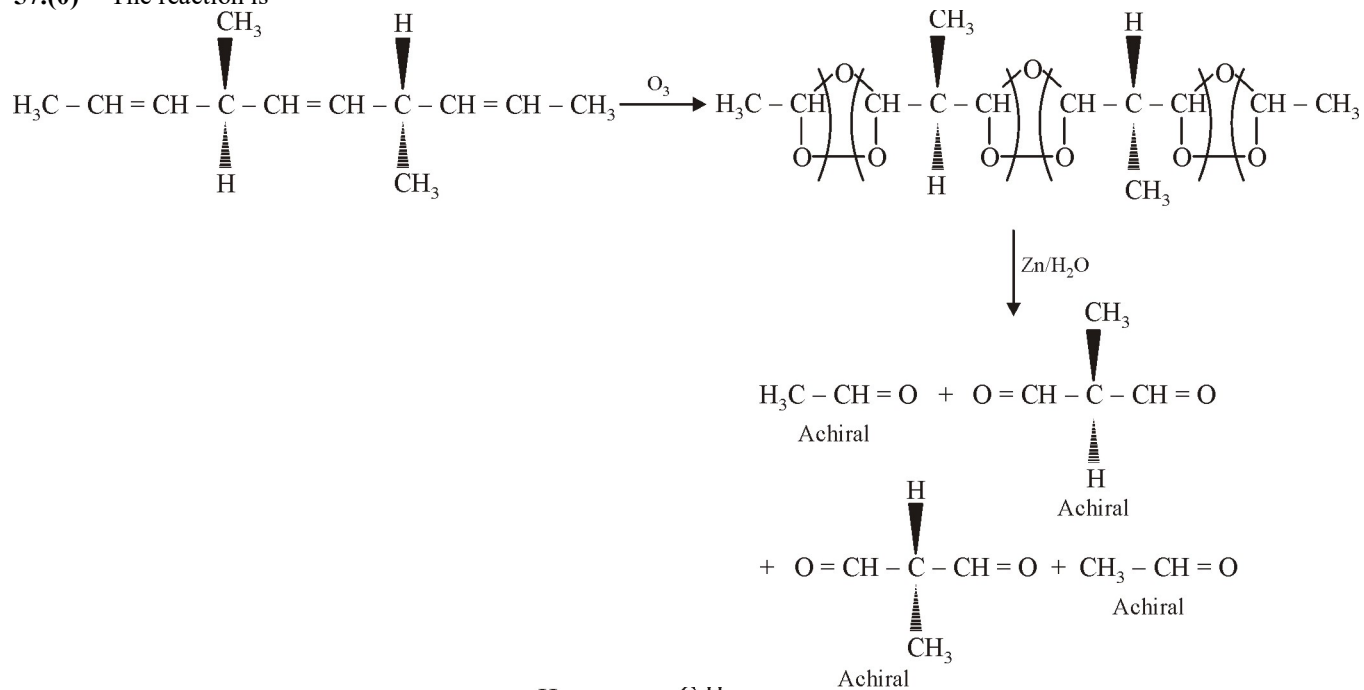
Hint : Symmetric Trans compound on anti-addition gives meso compound.

56.(8)

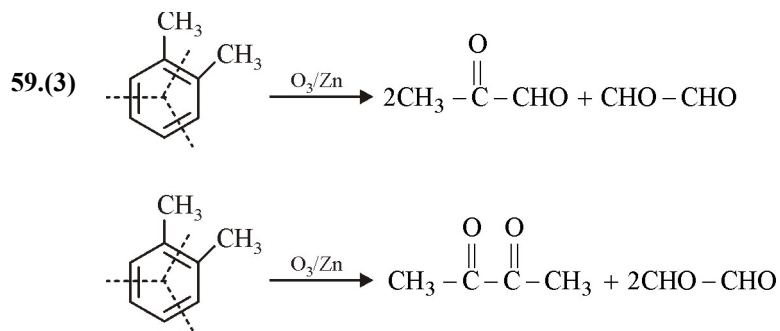


$$\text{Total} = 2 + 4 + 1 + 1 = 8$$

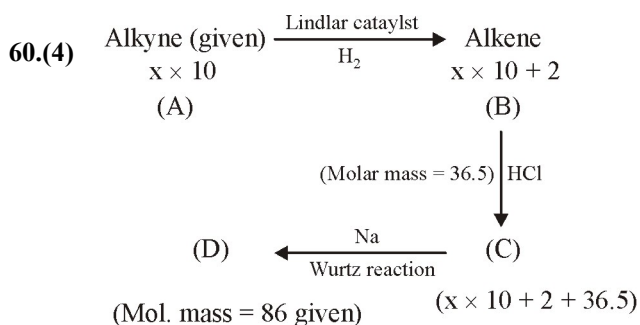
57.(0) The reaction is



Total 3 trans alkene can be formed.



3 different product can be formed



Calculated mass of (D) = $2(10 \times x + 2 + 36.5) - 2 \times 35.5 = 20x + 6$

(After Wurtz reaction, deduct two chlorine atoms)

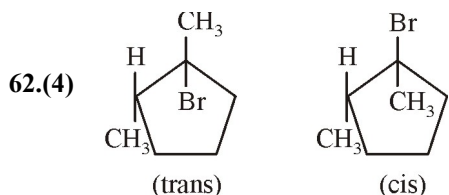
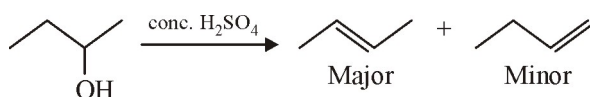
Therefore, Mol. mass calculated = Mol. Mass given

$20x + 6 = 86$

$x = \frac{86 - 6}{20} = 4$

Therefore, the alkyne is propyne with molecular mass 40.

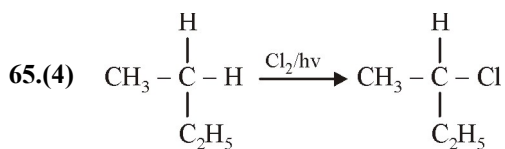
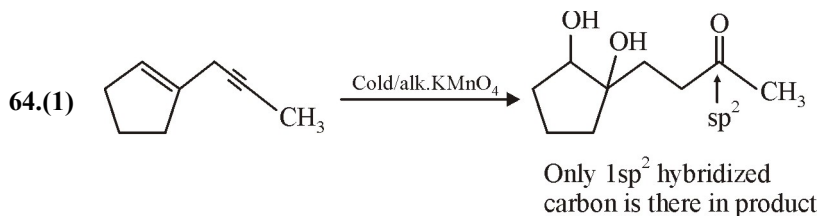
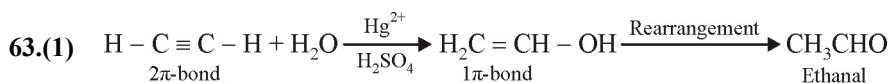
- 61.(3) The total number of possible products is three 2-butene can exist in cis form as well as trans form so total 3 product could be possible.



Since both of them lack symmetry both have enantiomeric pair

Hence total no. of isomers are 4.

Note: Only 3° product is formed because Br₂ is highly selective.

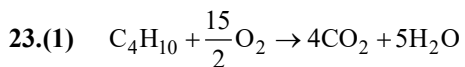


Minimum 4 carbons are required

Environmental Chemistry

21.(5) Water is considered as pure if its BOD is less than 5 ppm.

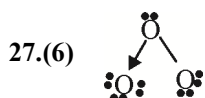
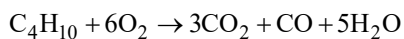
22.(5) Fe is co-ordinated to 5 groups and 6th site is free to which CO links if present.



24.(3)

25.(4)

26.(8)



28.(5)

29.(5) CO₂ and other green house gases are methane, water vapour, nitrous oxide, CFC's and ozone.

30.(5)

31.(7)

32.(1) Zn = 5ppm
Fe = 0.2 ppm
= (5 × 0.2) = 1

33.(4) A-Hydroxyapatite $[3Ca_3(PO_4)_2 \cdot Ca(OH)_2]$

B-harder fluorapatite $[3(Ca_3PO_4) \cdot CaF_2]$

$|A - B| = 17(2) - 19(2) = 4$

